

# The Symmetry Status

## Kernel Necessity and Independent Role Occupation

A fixed-domain normal form for Standard Model symmetry claims

### Publication theorem

For every role-realization-complete certificate  $C$  and fixed benchmark target  $T$ , independently formed target obligations and physical residues determine one lawful assignment orbit:

$$|\text{Assign}(C, T) / \Gamma_{\text{lawful}}(C, T)| = 1.$$

The public symmetry-status record is the canonical presentation of that unique orbit.

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Research article with controlled atlas, evidence archive, and reproducible audit package

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# Abstract

Quantum field theory uses the word *symmetry* for structures that do different work: constraints on admissible dynamics, equivalences of representatives, actions on physical states and operators, theorem consequences, anomaly or matching obstructions, parameter-limit regularities, target data, and transports on theory space. A single exact/broken or fundamental/derivative label cannot preserve those distinctions. This article develops a fixed-domain calculus that types each function separately and proves a unique role-and-status normal form for a deliberately difficult Standard Model benchmark universe.

AASC is not an optional interpretation added to independently determinate physics. A symmetry claim can function as a fixed, same-domain, consequence-bearing theorem object only if target identity, admissible comparison, standing-bearing use, lawful transformation, and irreversible act-time fixation are already available. Those conditions instantiate the Admissibility–Standing–Reference–Irreversibility kernel. Conventional quantum-field-theoretic formulas and source theorems supply the public sector-specific realization inside that prior standing-bearing domain; they do not create the domain’s determinacy from outside it.

The classified object is an atomic target-indexed claim of kind action, structure, diagnostic, or transport. Its physical disposition and its classical, quantum, anomaly, vacuum, boundary/global-form, operator, scale, and causal-ancestry coordinates are typed independently. The role-occupation theorem is instantiated with logically independent inputs. Target packets force structural role positions without naming their occupants. Physical and dependency facts generate candidate residues before any role-position lookup. Every role-position/residue pair is evaluated; complete matchings are tested by family-specific joint compatibility; and assignments are quotiented by the full incidence automorphism group preserving the theorem-bearing invariant tuple.

Across forty-six atomic records organized into sixteen maximum-stress benchmark families, the independent construction yields sixty-eight role positions, sixty-eight presentation-quotiented physical residues, ninety eligible incidence edges, twenty-nine complete preassignments, seven substantive compatibility rejections, twenty-two compatible assignments, and exactly one lawful assignment orbit per family. Gate uniqueness and apex admissible-interior uniqueness enter only after this local singleton theorem, excluding a residual second selector or governance-non-equivalent regime. The conclusion establishes classifier adequacy and unique benchmark normal form; it does not claim numerical parameter closure, arbitrary ultraviolet uniqueness, or the full candidate-exhaustion result reserved for SYM-11.

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## Read this first: dependency and proof spine

$$\text{DetSymClaim}(S, D) \implies K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) \implies A_D^+.$$

$A_D^+$  + fixed physical carrier package  $\implies$  conventional-facing QFT specialization,

$$T \implies \mathcal{R}_T \quad \text{without occupant identity,}$$

$$T + C_{\text{phys}} \implies X_T / \sim_{\text{pres}} \quad \text{without role lookup,}$$

$$(\mathcal{R}_T, X_T, C) \implies E_C, J_C, \Gamma_{\text{lawful}},$$

$$\implies |\text{Assign}(C, T) / \Gamma_{\text{lawful}}| = 1,$$

$$\implies \text{no residual same-domain selector or regime plurality.}$$

Within each record, the public QFT certificate is completed before the final AASC role is reported. That methodological order prevents role labels from manufacturing their premises. It does not reverse the logical dependency: determinate physical use is already kernel-bearing.

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# Dependency Direction and Proof-Class Lock

## 1.1 AASC is necessity, not an optional overlay

A symmetry claim becomes available for determinate use only when the same target, acting object, acted-upon object, transformation, comparison class, quotient, consequence, scope, and reportable endpoint persist through the classification act. Those are not optional interpretive conveniences. They instantiate the kernel

$$K_D = (\text{Admissibility, Standing, Reference, Irreversibility})$$

before a conventional field-theoretic representative can carry theorem force [12, 19]. The governing implication is therefore

$$\text{DetSymClaim}(S, D) \implies K_D \implies A_D^+.$$

Ordinary quantum field theory remains indispensable: it supplies explicit transformation laws, invariant observables, currents, constraints, anomaly statements, vacuum realizations, boundary qualifications, operator loci, and scale ranges. Its role here is faithful specialization and public physical audit. It is not a standing-neutral exterior from which target identity or admissibility is first created.

### Blocked dependency inversion

The project does not assume an independently complete physical ontology and then append an AASC vocabulary. It derives the conditions under which a determinate physical specialization can be fixed and used at all, and then checks that the Standard Model specialization is faithful to those conditions.

## 1.2 Kernel cost of denial

The necessity claim is operationally testable through denial costs.

| Kernel role denied | Consequence for symmetry classification                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| Reference          | No fixed target, acting object, quotient, residue identity, comparison class, or same-scope rival relation remains.  |
| Standing           | A transformation or invariant remains notation without determinate downstream theorem use, import, or reportability. |

| Kernel role denied | Consequence for symmetry classification                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|
| Admissibility      | No boundary separates lawful redescription, physical action, repair, selector import, hidden load, or target change.     |
| Irreversibility    | Later anomaly, vacuum, boundary, or scale information can retrospectively rewrite the same completed classification act. |

A critic who retains the same fixed symmetry target, comparison relation, and theorem-bearing consequence while denying the kernel has retained the kernel functions under another name. A critic who removes those functions has changed or destroyed the target. There is no neutral exterior position at the same scope.

### 1.3 Proof class and exact burden

This article is a kernel-forced, fixed-domain, finite role-occupation theorem with imported QFT certificates. It is not a derivation of the Standard Model from empty premises and not a replacement for Noether theory, BRST cohomology, anomaly theory, vacuum analysis, generalized symmetry, or renormalization-group calculation. Those physical results are imported with exact targets and source locations. The local theorem establishes that, once the complete certificate is fixed, the finite role-assignment problem has one lawful orbit.

A criticism has standing if it does at least one of the following: breaks kernel necessity while preserving the same determinate target; invalidates an imported theorem or its specialization; identifies an omitted target-relevant residue or role obligation; breaks an eligibility or compatibility edge; shows that the presentation quotient is too coarse or too fine; exposes an unaccounted lawful automorphism; or supplies a second complete assignment orbit at the same target. Preference for another vocabulary, deletion of a theorem premise, demand that software independently rediscover imported physics, or change of target is not a same-scope counterexample.

### 1.4 Two coordinated ledgers

The project maintains two synchronized ledgers.

1. The **conventional-facing physical ledger** records target, action, invariant, quotient, current or constraint, anomaly, vacuum, boundary/global form, operator locus, and scale. It can be checked without accepting AASC terminology.
2. The **kernel-faithful role ledger** records which target obligation each independently formed residue can occupy and why competing occupations fail.

The first ledger is completed before final role reporting, preventing circular role assignment. The kernel remains logically prior because the first ledger's determinate identity, target, comparison, and standing-bearing use already presuppose it.

# Fixed Domains and the Three Uniqueness Layers

## 2.1 Target fixation

A benchmark target fixes every coordinate capable of changing the theorem object: field content, interactions, background, boundary, topology, faithful global form, operator category, phase, neutrino convention, perturbative or nonperturbative regime, and scale or EFT interval. A change in one of those coordinates is a target update. It is not a same-act repair and cannot be presented as a rival inside the source target.

Let  $T$  denote a fixed atomic benchmark target and let  $C$  denote its complete certificate. The classification domain is

$$D_{C,T} = (E_{C,T}, A_{C,T}, F_{C,T}, G_{C,T}, \text{DAC}_{C,T}),$$

where  $E_{C,T}$  is the finite output envelope,  $A_{C,T}$  the lawful transformation class,  $F_{C,T}$  the fixation packet,  $G_{C,T}$  the complete classification gate, and  $\text{DAC}_{C,T}$  closure under lawful continuation. Target identity includes certificate version and classifier-rule version, so silent source or rule drift cannot masquerade as a same-domain result.

## 2.2 Gate, assignment, and interior uniqueness

Three uniqueness claims are distinct and enter in a strict order.

**U1. Gate uniqueness.** Two fixed-domain admissibility gates cannot disagree extensionally; a second primitive authorizer either coincides with the gate, changes target, or is inadmissible [23].

**U2. Assignment uniqueness.** The local SYM-0 construction proves

$$|\text{Assign}(C, T) / \Gamma_{\text{lawful}}(C, T)| = 1.$$

This is the theorem that supplies uniqueness of the individual role-and-status normal form.

**U3. Interior uniqueness.** After local singleton occupation, apex closure excludes a residual governance-non-equivalent admissible interior or selector [12, 23].

No inference from one maximal admissible interior to one element is used. The local assignment theorem carries the element-level burden; the apex result closes the regime after that burden is discharged.

## 2.3 The finite assignment object

For a fixed family target, let

$$\mathcal{R}_T = \{r_1, \dots, r_n\}$$

be the role positions forced by target obligations, and let

$$X_T = X_T^{\text{raw}} / \sim_{\text{pres}}$$

be the independently generated physical residues after lawful presentation quotienting. The role positions contain structural functions, level, locus, forcing basis, and compatibility obligations; they contain no expected occupant, candidate name, source-record ID, or semantic subject. The residues contain physical and dependency facts, invariant content, quotient status, level, locus, and failure behavior; they contain no final structural role or role alias.

The eligibility relation is

$$E_C \subseteq \mathcal{R}_T \times X_T.$$

A complete assignment is a bijection  $\sigma : \mathcal{R}_T \rightarrow X_T$  satisfying every pointwise eligibility edge and the family-level joint-compatibility predicate  $J_C(\sigma) = 1$ . The lawful group  $\Gamma_{\text{lawful}}$  consists of automorphisms of the role architecture and residue invariant structure that preserve incidence and compatibility. The theorem-bearing object is the orbit space, not a stored database row.

## 2.4 Why the inputs are independent

Independence is enforced at construction time.

- The target role architecture is derived before any residue is inspected and contains an empty occupant-identity field.
- Candidate residues are generated from candidate kind, physical disposition, status, quotient, dependency, transition, regime, and failure facts before the role architecture is loaded.
- Every same-family role-position/residue pair is evaluated. No expected name, exact semantic subject, source-record identity, or role alias is used as an eligibility test.
- Joint compatibility rejects complete pointwise-eligible matchings when their causal orientation, co-instantiation, transition, regime, failure, or same-locus constraints conflict.
- Automorphism signatures exclude source-record IDs, candidate display names, slot ordinals, and expected occupants.

These conditions close the possibility that the conclusion enters upstream through role-specific residue formation or occupant-bearing slot identity.

# Atomic Claims, Evidence, and Independent Residue Formation

## 3.1 Four controlled layers

The canonical project separates four objects that prose often conflates.

1. **Atomic claim.** Record identity, family, fixed target, candidate kind, acting or structural object, acted-upon object, law, and operator locus.
2. **Admitted facts.** Source-bound, target-matched propositions about invariance, quotient behavior, currents, constraints, anomalies, vacuum realization, boundaries, global form, scale, removal, dependency, transition, and obstruction.
3. **Rules.** Physical-status extraction, residue formation, role eligibility, joint compatibility, support, completion, nonclaim, and contradiction rules.
4. **Generated results.** Candidate residues, assignment orbits, public status records, support packets, completion dispositions, and nonclaims.

Generated conclusions are never accepted as raw premises.

## 3.2 Fact admission

A fact  $f$  is admitted only when

$$\begin{aligned} &\text{SourceBound}(f) \wedge \text{TargetMatched}(f) \\ &\wedge \text{DerivationLicensed}(f) \wedge \text{VersionFixed}(f) \wedge \text{ScopeComplete}(f). \end{aligned}$$

Each fact identifies one or more proposition-level source claims, exact source location, target assumptions, rule version, and content digest. Citation metadata and theorem import remain distinct: the project verifies bibliographic identity and certificate linkage, while the truth of an imported QFT theorem remains an explicit premise rather than a software-generated result.

## 3.3 Candidate kinds

The top-level sum type is

$$\text{CandidateKind} = \text{Action} \sqcup \text{Structure} \sqcup \text{Diagnostic} \sqcup \text{Transport}.$$

The physical disposition is indexed by kind. Actions may be quotient-trivial, quotient-nontrivial, or cohomological; structures are target or architectural data; diagnostics are consequences or

obstructions rather than actions; transports are representative redescriptions or physical theory-space relations. This prevents the trace anomaly, line spectrum, global form, or RG flow from being forced into an ordinary state-action vocabulary.

### 3.4 Atomic decomposition

Familiar names are parent families, not automatically atomic classifier objects. The gauge family separates architecture, compact-support operation, boundary action, large transformation, topology, anomaly, and global form. Poincaré separates domain structure, active action, passive coordinates, and Noether consequence. Flavor separates limit symmetry, basis change, spurion organization, invariant mixing data, and target-changing neutrino assumptions. Scale separates coordinate transformation, representative scale choice, physical RG flow, matching, fixed-point action, and trace diagnostic.

SYM-0 proves decomposition for the declared sixteen-family calibration universe. It does not claim full Standard Model candidate exhaustion; that global generation burden remains assigned to SYM-11. Within the declared universe, every registered parent has a fixed child partition and every child generates at least one physical residue before role assignment.

### 3.5 Presentation quotient and residue identity

Raw candidate facets are identified only when a lawful presentation transformation preserves target, physical quotient, invariant observables, status ancestry, continuation, failure conditions, and reportable scope. The quotient

$$X_T = X_T^{\text{raw}} / \sim_{\text{pres}}$$

removes gauge, basis, coordinate, scheme, or line-presentation multiplicity without merging physically distinct actions, target data, anomalies, phases, or scale regimes. A residue's source-record trace is retained for provenance but excluded from the lawful automorphism signature.

## Physical Status and Causal Ancestry

### 4.1 Kind-indexed physical disposition

For an atomic claim  $S$  in target  $D$ , the physical packet is

$$\text{SymStat}_D(S) = (Q, C, Q_m, V, B, G_{\text{op}}, R, A, E).$$

Here  $Q$  is physical disposition;  $C$  classical status;  $Q_m$  quantum status;  $V$  vacuum realization;  $B$  boundary, topology, and global-form dependence;  $G_{\text{op}}$  operator locus;  $R$  scale or EFT range;  $A$  anomaly type; and  $E$  evidence and completion status. Each coordinate is independently generated from admitted facts.

### 4.2 Status-coordinate non-collapse

Exactness, physicality, anomaly, vacuum realization, scale range, and structural standing are not mutually inferable. CPT may be exact and theorem-derived. A compact-support gauge operation may be quotient-trivial while its licensing architecture is load-bearing. Isospin may be approximate and physically active. A 't Hooft anomaly may certify robust matching rather than erase the global symmetry. A fixed global form may be target data rather than a transformation. Benchmark witness pairs establish that no ordinary coordinate projection determines the complete normal form.

### 4.3 Quotient discipline

For an action  $g$  on  $X_D$  with quotient  $q_D : X_D \rightarrow X_D^{\text{phys}}$ ,

$$q_D(gx) = q_D(x) \quad \forall x$$

defines quotient-triviality, while a witness  $x$  satisfying  $q_D(gx) \neq q_D(x)$  establishes a physical action. Apparent mixed cases are parent families whose atomic loci have different statuses; “mixed” is not an atomic escape value. Structures, diagnostics, and transports use their own typed disposition branches.

### 4.4 Causal ancestry

The calculus separates explicit deformation, anomaly, nonperturbative topology, vacuum selection, Higgs fixation, mass or coupling deformation, boundary condition, higher-dimension operator, RG departure, regulator artifact, and target update. Realization conditions are not

automatically failure causes, and target differences are not symmetry breaking. Two non-exact records are equivalent only when their target, locus, status vector, and causal ancestry agree.

## 4.5 Completion and fail-closed disposition

Schema completeness, certificate completeness, physical resolution, and role closure are independent. A submitted claim routes to one of the declared dispositions: normal form, conditional rule, incomplete certificate, incomplete target, open burden, target update, grammar failure, rejection, or supersession. Missing evidence never defaults to a role or to normal form. Removing a load-bearing premise produces an incomplete certificate; it does not create a rival theorem.

# Independent Role Occupation and Singleton Closure

## 5.1 Target-derived role positions

The structural vocabulary is

$$\mathcal{R} = \{\text{Anchor}, \text{CarrierConstraint}, \text{BridgeFixation}, \text{PhysicalAction}, \\ \text{DerivedWitness}, \text{Skin}, \text{EffectiveRoute}, \text{Obstruction}, \text{TargetUpdate}\}.$$

A benchmark target forces a finite multiset of role positions. A position specifies the structural function to be occupied, its target level and locus, its forcing basis, and its compatibility obligations. It does not specify a candidate, source record, canonical name, semantic subject, or expected occupant. Multiple positions of one role type are lawful when distinct target obligations require them.

## 5.2 Physical residues formed before role lookup

Candidate facets are generated from physical and dependency facts alone. Examples include target definition, architecture dependency, transition mediation, target comparison, structural consequence, action object, presentation relation, cohomological relation, diagnostic content, regime condition, consistency condition, transport relation, and structural object. Those facet names describe physical relations, not structural roles. Residues are presentation-quotiented before the role architecture is consulted and remain present whether or not they are eligible for any role.

## 5.3 Eligibility incidence

Only after the two finite sets are fixed is the relation

$$E_C \subseteq \mathcal{R}_T \times X_T$$

evaluated. A pair survives when the residue supplies the capabilities required by the target obligation, preserves target fixation, has the correct causal orientation, and carries the necessary positive or eliminative certificates. Every same-family pair is recorded. Exact candidate names, source IDs, semantic-subject equality, and role aliases are forbidden as eligibility predicates.

The resulting graph is intentionally nontrivial. Twenty role positions admit more than one residue, and nineteen residues are eligible for more than one position. This competition is resolved downstream; it is not suppressed during formation.

## 5.4 Joint compatibility

Pointwise eligibility is necessary but insufficient. A complete matching must also satisfy

$$J_C(\sigma) = 1.$$

The predicate checks bijection and common target, causal orientation, distinct physical facets for multi-level occupation, same-locus role exclusions, regime-to-licensed-structure coherence, failure-to-theorem-bearing-subject coherence, and transition-to-architecture coherence. In the frozen benchmark universe, twenty-nine complete pointwise-eligible preassignments are found; seven are rejected by joint compatibility, leaving twenty-two compatible assignments before quotienting.

## 5.5 Lawful automorphisms

The group  $\Gamma_{\text{lawful}}$  is computed from the invariant role and residue structures. Slot signatures exclude slot ordinals and occupant identity. Residue signatures exclude source-record IDs and display names. Permutations are retained only when they preserve the full eligibility relation and joint compatibility. The quotient therefore collapses lawful presentation or incidence symmetries without using over-individuating database labels to force the identity group.

## 5.6 Assignment theorem

Define

$$\text{Assign}(C, T) = \left\{ \sigma : \mathcal{R}_T \xrightarrow{\sim} X_T : E_C(r, \sigma(r)) \ \forall r, \ J_C(\sigma) = 1 \right\}.$$

The unpruned family spaces contain 39,927,933 mathematical bijections. Sound eligibility pruning visits 124 search nodes, exposes all twenty-nine complete preassignments, and applies  $J_C$  to each. The twenty-two compatible assignments form exactly sixteen lawful orbits, one per family.

**Theorem 5.1** (Independent singleton occupation). For every role-realization-complete certificate  $C$  and fixed benchmark target  $T$  in the declared SYM-0 universe,

$$|\text{Assign}(C, T) / \Gamma_{\text{lawful}}(C, T)| = 1.$$

*Proof.* Role positions and residues are finite and independently formed. Every incidence pair is evaluated, so the eligibility graph is complete relative to the frozen certificate. Backtracking enumerates every perfect matching of that graph. The family-specific predicate  $J_C$  rejects every incompatible complete matching. The computed invariant-preserving automorphism group acts on all compatible assignments; orbit decomposition leaves one orbit. The proof object for each family records the role positions, residues, eligible edges, rejected preassignments, compatible assignments, group elements, and canonical orbit representative.  $\square$

## 5.7 Role-realization completeness

A certificate is complete only when it includes evidence for the declared benchmark child partition, all-and-only presentation quotient, target-derived role architecture, independent residue formation, complete all-pair eligibility, complete compatibility evaluation, failure-route coverage, and lawful orbit decomposition. A Boolean declaration is not accepted as evidence. Deleting a

required item produces `IncompleteCertificate`; it cannot silently change the unique occupation.

## Redescription, Support, and Rivals

### 6.1 Redescription covariance

A lawful redescription preserves target identity, standing, quotient content, invariant physical data, status ancestry, continuation, failure conditions, and reportable consequences. It acts on presentations and may induce permutations of residues or assignments, but it cannot alter the lawful orbit. Formally,

$$\gamma(E_C) = E_C, \quad J_C(\sigma) = J_C(\gamma \cdot \sigma), \quad [\gamma \cdot \sigma] = [\sigma].$$

The general covariance result follows from the fixed-domain quotient and role-occupancy machinery [13, 15, 20]. Six executed regressions provide concrete witnesses: proper gauge covariance, CKM rephasing, passive Lorentz transformation, scheme reparameterization, scale-coordinate change, and integral line-charge basis change. Each calculation verifies the relevant invariant and then confirms unchanged occupation projection. These regressions audit the specialization; they are not substitutes for the general theorem.

### 6.2 Carrier-support dependencies

The support graph records theorem-indexed relations  $K_i \chi_T S_j$ . Each retained edge identifies its source carrier or theorem, the facts it supports, the target coordinates held fixed, and the consequence of denial. The claimed result is relative: the frozen graph is integrity-checked and inclusion-minimal relative to the imported theorem set and declared dependency relation. No claim of a metaphysically unique explanatory graph is made. Full alternative-theorem discovery remains a corpus governance burden rather than a task delegated to graph syntax.

### 6.3 Rival disposition

A proposed rival must state the same target, role architecture, candidate residues, quotient, eligibility evidence, compatibility evidence, and lawful transformation class. It then falls into one of the following classes: the same lawful orbit; skin-only change; ineligible assignment; jointly incompatible assignment; same-act repair; hidden selector import; target update; incomplete certificate; or genuine counterexample. A genuine counterexample is a second complete assignment orbit preserving all fixed-domain conditions.

### 6.4 Standing-filtered criticism

An objection has theorem standing if it breaks kernel necessity while preserving the same determinate object, invalidates an imported theorem or its specialization, exposes an omitted role

obligation or physical residue, breaks the quotient, eligibility, compatibility, automorphism, or support relation, or supplies a faithful second orbit. Premise deletion, unsupported fact insertion, software convention, changed target, and journal preference do not by themselves refute the theorem. Audit defects remain important and are repaired fail-closed, but they are not automatically same-scope physical counterexamples.

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## Benchmark dossiers I: gauge, constraints, and spacetime

The benchmark dossiers are generated from the canonical atlas. Record identity, target, generated status, role residue, support packet, and source claims therefore cannot drift independently of the controlled data.

### 7.1 Benchmark 1: Compact-support local gauge transformation

**Family:** FAM-01-GAUGE-PROPER. **Partition certificate:** PART5-FAM-01-GAUGE-PROPER.

Architecture and proper operation are distinct; the family covers the load-bearing gauge architecture and its quotient-trivial compact-support operation.

Children are disjoint by candidate kind and logical level.

**Principal sources:**

`citeHenneauxTeitelboim1992;`

`citeMaleySM2.`

### 7.1.1 Standard Model gauge architecture

**Record:** SYM-00-GAUGE-ARCH-v5 **Target:** D-SM-CORE **Constructor:** CTR-TARGET-ARCH

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Architecture/Target.

**Object and law:** Gauge-product, transition, connection, representation, charge, and anomaly-cancellation architecture. Covariant derivative, bundle transition, representation, charge, and anomaly-cancellation compatibility

**Acted upon:** Admissible field configurations and quantum measure.

**Invariant or preserved content:** Gauge-covariant dynamics and gauge-invariant observable algebra.

**Physical projection:** Architecture licenses the gauge-orbit/BRST physical quotient.

**Generated disposition:** StructuralDatum.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Not Applicable**; scale **AllTargetScales** (Renormalizable target with running couplings treated by separate transport records). **Operator and global scope.** Operator loci: Fields, LocalOperators, States. Boundary and global dependence: global-form-sensitive; insensitive within the target to boundary-sensitive, topology-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: AnomalyFreeArchitecture, TrivialBundleSector. Target differences: none. **Role residue.** CarrierConstraint [Architecture/Target]. **Disposition and resolution.** Normal Form; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-ARCH-v5-01, CONTRIB-EDGE5-SYM-00-GAUGE-ARCH-v5-02, CONTRIB-EDGE5-SYM-00-GAUGE-ARCH-v5-03. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.1.2 Compact-support proper gauge operation

**Record:** SYM-00-GAUGE-LOCAL-PROPER-v5 **Target:** D-SM-CORE **Constructor:** CTR-PROPER-REDESCRIPTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; Representative/Bulk.

**Object and law:** Smooth gauge map connected to the identity and equal to the identity at the admitted boundary. A maps to  $gAg^{-1}-(i/g)(dg)g^{-1}$ ; matter fields map by their gauge representations

**Acted upon:** Gauge and matter field representatives.

**Invariant or preserved content:** Gauge-invariant observables and physical cohomology class.

**Physical projection:** q maps representatives to gauge orbits/BRST classes after proper-gauge identification.

**Generated disposition:** TrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Not Applicable**; scale **AllTargetScales** (Anomaly-free core target and fixed proper-gauge locus). **Operator and global scope.** Operator loci: Fields, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: AnomalyFreeArchitecture, HiggsFixedPhase, TrivialBundleSector. Target differences: none. **Role residue.** Skin [Representative/Bulk]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-LOCAL-PROPER-v5-01, CONTRIB-EDGE5-SYM-00-GAUGE-LOCAL-PROPER-v5-02. **Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness

under arbitrary ultraviolet premises.

## 7.2 Benchmark 2: Gauge transformation nontrivial at a boundary

**Family:** FAM-02-GAUGE-BOUNDARY. **Partition certificate:** PART5-FAM-02-GAUGE-BOUNDARY.

The family separates the abstract boundary-charge rule from one instantiated asymptotic  $U(1)$  action.

The rule is diagnostic/conditional; the example is an action in a fixed boundary target.

**Principal sources:**

citeHenneauxTeitelboim1992;

citeMaleySM2.

### 7.2.1 Boundary gauge-action conditional rule

**Record:** SYM-00-GAUGE-BOUNDARY-RULE-v5 **Target:** D-SM-BOUNDARY-RULE **Constructor:** CTR-BOUNDARY-RULE

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Boundary/Boundary.

**Object and law:** Gauge-like transformation with admitted nonzero boundary value. The differentiable generator equals the bulk constraint plus a surface term  $Q[\alpha]$

**Acted upon:** Bulk representatives together with boundary data or edge modes.

**Invariant or preserved content:** Bulk gauge-invariant observables; boundary data transform when  $Q[\alpha]$  is finite and nonzero.

**Physical projection:** Only zero-charge proper transformations are divided out.

**Generated disposition:** ConditionalActionRule.

**Generated status.** Classical Exact; quantum Conditional; anomaly Unresolved; vacuum StateDependent; scale NotApplicable (Conditional theorem over boundary-completed targets). **Operator and global scope.** Operator loci: States. Boundary and global dependence: boundary-sensitive, topology-sensitive, global-form-sensitive; insensitive within the target to line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: NonzeroBoundaryParameter, IntegrableSurfaceCharge. Target differences: none. **Role residue.** DerivedWitness [Consequence/Boundary]. **Disposition and resolution.** ConditionalRule; record completeness Complete; physics resolution Conditional. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-BOUNDARY-RULE-v5-01, CONTRIB-EDGE5-SYM-00-GAUGE-BOUNDARY-RULE-v5-02. **Rival disposition.** Treating the conditional boundary theorem as an already instantiated physical action. The record is a Diagnostic/ConditionalActionRule; only the asymptotic U(1) child is CompleteInstantiated.

#### Exact nonclaims

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

The record is structurally classified but its unresolved physics is not reported as exact.

### 7.2.2 Asymptotic $U(1)$ transformation generated by electric charge

**Record:** SYM-00-U1-ASYMPTOTIC-CHARGE-v5 **Target:** D-U1-ASYMPTOTIC **Constructor:** CTR-PHYSICAL-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; Boundary/Boundary.

**Object and law:** Asymptotically constant  $U(1)$  gauge parameter. The improved generator is the Gauss constraint plus  $\alpha_{\infty}$  times the electric flux at infinity

**Acted upon:** Charged asymptotic states and boundary electric flux.

**Invariant or preserved content:** Bulk gauge-invariant dynamics while charged states acquire the global phase.

**Physical projection:** Transformations with zero asymptotic charge are quotiented; the constant boundary mode survives.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Fixed anomaly-free  $U(1)$  target and falloff class). **Operator and global scope.** Operator loci: States. Boundary and global dependence: boundary-sensitive; insensitive within the target to topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: NonzeroBoundaryParameter, IntegrableSurfaceCharge. Target differences: none. **Role residue.** PhysicalAction [Boundary/Boundary]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U1-ASYMPTOTIC-CHARGE-v5-01, CONTRIB-EDGE5-SYM-00-U1-ASYMPTOTIC-CHARGE-v5-02.

**Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 7.3 Benchmark 3: Large gauge transformation and theta sector

**Family:** FAM-03-GAUGE-LARGE. **Partition certificate:** PART5-FAM-03-GAUGE-LARGE.

The family partitions large/topological gauge material into quotiented action, theta-represented action, target update, in-target topology anchor, and global-anomaly obstruction.

Children differ by target, disposition, or candidate kind; controlled overlap is only dependency.

**Principal sources:**

citeHenneauxTeitelboim1992;

citeMaleySM2;

citeWitten1982;

citeMaleyAER.

### 7.3.1 Large transformation included in the gauge quotient

**Record:** SYM-00-GAUGE-LARGE-QUOTIENT-v5 **Target:** D-YM-LARGE-QUOTIENT **Constructor:** CTR-LARGE-QUOTIENT

#### Atomic claim and generated disposition

**Kind and locus:** Action; Representative/Topology.

**Object and law:** A specified disconnected gauge transformation declared proper in the target. Standard gauge law in the selected winding class

**Acted upon:** Field representatives and wave functionals.

**Invariant or preserved content:** All admitted observables after the enlarged quotient.

**Physical projection:** The selected disconnected component is included in the equivalence relation.

**Generated disposition:** TrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State Dependent**; scale **TopologicalSector** (Fixed component and quotient convention). **Operator and global scope.** Operator loci: Fields, States. Boundary and global dependence: topology-sensitive, global-form-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, line-spectrum-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: **NontrivialBundleSector**. Target differences: none. **Role residue.** Skin [Representative/Topology]. **Disposition and resolution.** **NormalForm**; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-LARGE-QUOTIENT-v5-01, CONTRIB-EDGE5-SYM-00-GAUGE-LARGE-QUOTIENT-v5-02.

**Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.3.2 Large transformation represented on a theta sector

**Record:** SYM-00-GAUGE-LARGE-THETA-ACTION-v5 **Target:** D-YM-THETA-SECTOR **Constructor:** CTR-PHYSICAL-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Unit-winding large gauge transformation.  $\Psi_{\theta}[A^g] = \exp(i \theta \int n(g)) \Psi_{\theta}[A]$

**Acted upon:** Theta-vacuum wave functionals.

**Invariant or preserved content:** Local gauge-invariant observables; theta-sector representation data are physical.

**Physical projection:** The disconnected operation is represented rather than divided out.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State Dependent**; scale **TopologicalSector** (Fixed theta representation). **Operator and global scope.** Operator loci: States, PartitionFunction. Boundary and global dependence: topology-sensitive, global-form-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, line-spectrum-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: NontrivialBundleSector, ThetaSectorSelected. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-LARGE-THETA-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-GAUGE-LARGE-THETA-ACTION-v5-02. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

The conclusion is restricted to the explicitly named controlled target variant.  
The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.3.3 Fixed bundle and topological-sector datum

**Record:** SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5 **Target:** D-SM-TOPOLOGY-Z6 **Constructor:** CTR-DOMAIN-ANCHOR

#### Atomic claim and generated disposition

**Kind and locus:** Structure; TargetData/Target.

**Object and law:** Fixed principal-bundle class, instanton number, and admitted topological sectors. The path integral and operator category are indexed by the selected global form and bundle sectors.

**Acted upon:** Partition function, state sectors, and extended operator spectrum.

**Invariant or preserved content:** The selected topological sector and its partition-function weighting.

**Physical projection:** Not an operation; the datum fixes which configurations and sectors belong to the target..

**Generated disposition:** StructuralDatum.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State Dependent**; scale **TopologicalSector** (Fixed global/topological target). **Operator and global scope.** Operator loci: **PartitionFunction**, **ExtendedOperators**. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: **NontrivialBundleSector**, **FixedGlobalFormZ6**, **ExtendedOperatorsAdmitted**. Target differences: none. **Role residue.** Anchor [TargetTyping/Target]. **Disposition and resolution.** **NormalForm**; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5-02, CONTRIB-EDGE5-SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5-03. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.3.4 Admission of nontrivial gauge topology as a target update

**Record:** SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 **Target:** D-SM-TOPOLOGY-UPDATE  
**Constructor:** CTR-TOPOLOGY-UPDATE

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Architecture/Topology.

**Object and law:** Bundle class, winding lattice, and theta periodicity. Topological classes and allowed bundles determine sector decomposition and theta identifications

**Acted upon:** Definition of the gauge theory and partition-function sector.

**Invariant or preserved content:** Global theory identity and extended-operator content.

**Physical projection:** Topological data are target structure, not an operation.

**Generated disposition:** TargetUpdateDisposition.

**Generated status.** Classical NotApplicable; quantum NotApplicable; anomaly NotApplicable; vacuum NotApplicable; scale TopologicalSector (Fixed global/topological target). **Operator and global scope.** Operator loci: PartitionFunction, ExtendedOperators. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: GlobalForm, OperatorCategory, Topology. **Role residue.** TargetUpdate [TargetComparison/Target]. **Disposition and resolution.** TargetUpdate; record completeness Complete; physics resolution NotApplicable. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.

- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 1 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5-01**. **Rival disposition.** Treating the changed target as a same-scope counterexample or internal role. The target-difference certificate routes it to TargetUpdate before same-scope role evaluation.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The changed target is not a same-scope rival to the source target.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.3.5 Witten SU(2) global-anomaly obstruction

**Record:** SYM-00-SU2-GLOBAL-ANOMALY-v5 **Target:** D-SU2-GLOBAL-ANOMALY-TEST **Constructor:** CTR-DIAGNOSTIC-OBSTRUCTION

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/Topology.

**Object and law:** Fermion determinant sign under the nontrivial  $\pi_4(\text{SU}(2))$  transformation. For an odd number of left-handed doublets the determinant changes sign

**Acted upon:** Quantum partition function.

**Invariant or preserved content:** Consistency of the quantum gauge measure.

**Physical projection:** No action quotient is assigned to the anomaly diagnostic.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical NotApplicable; quantum GaugeInconsistent; anomaly GlobalGauge; vacuum NotApplicable; scale TopologicalSector (Nonperturbative global consistency test). **Operator and global scope.** Operator loci: PartitionFunction. Boundary and global dependence: topology-sensitive, global-form-sensitive; insensitive within the target to boundary-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly. Realization conditions: NontrivialBundleSector. Target differences: none. **Role residue.** DerivedWitness [Consequence/Topology]; Obstruction [ConsistencyOrMatching/GlobalGauge]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [14, 26].

- **Certificate 1.** An  $\text{SU}(2)$  Anomaly; pp. 324-328; persistent authority 10.1016/0370-2693(82)90728-6. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** An  $\text{SU}(2)$  Anomaly; pp. 324-328; persistent authority 10.1016/0370-2693(82)90728-6. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** An  $\text{SU}(2)$  Anomaly; pp. 324-328; persistent authority 10.1016/0370-2693(82)90728-6. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** An  $SU(2)$  Anomaly; pp. 324-328; persistent authority 10.1016/0370-2693(82)90728-6. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-SU2-GLOBAL-ANOMALY-v5-01, CONTRIB-EDGE5-SYM-00-SU2-GLOBAL-ANOMALY-v5-02, CONTRIB-EDGE5-SYM-00-SU2-GLOBAL-ANOMALY-v5-03. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

#### Exact nonclaims

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 7.4 Benchmark 4: BRST symmetry

**Family:** FAM-04-BRST. **Partition certificate:** PART5-FAM-04-BRST.

The BRST family contains the cohomological differential/projection child required by the benchmark.

Single-child family in the present scope; ghost number is assigned to the dedicated gauge-sector analysis in SYM-1.

**Principal sources:**

citeHenneauxTeitelboim1992;

citeMaleySM2.

### 7.4.1 BRST differential and physical cohomology

**Record:** SYM-00-BRST-COHOMOLOGY-v5 **Target:** D-SM-CORE **Constructor:** CTR-COHOMOLOGICAL

#### Atomic claim and generated disposition

**Kind and locus:** Action; Diagnostic/FieldSpace.

**Object and law:** Nilpotent BRST differential  $s$ .  $s^2=0$ ; physical states/observables are cohomology classes

**Acted upon:** Gauge-fixed fields, ghosts, antifields, and local functionals.

**Invariant or preserved content:** Gauge-fixed quantum action and BRST cohomology.

**Physical projection:**  $\ker s / \operatorname{im} s$ .

**Generated disposition:** CohomologicalProjection.

**Generated status.** Classical **Exact**; quantum **PerturbativelyExact**; anomaly **None**; vacuum **NotApplicable**; scale **AllTargetScales** (Local perturbative BRST cohomology in the anomaly-free gauge-fixed formulation; nonperturbative global gauge-fixing questions are excluded.). **Operator and global scope.** Operator loci: Fields, LocalOperators, States. Boundary and global dependence: topology-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: GaugeFixingChosen, AnomalyFreeArchitecture. Target differences: none. **Role residue.** DerivedWitness [Consequence/FieldSpace]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-BRST-COHOMOLOGY-v5-01, CONTRIB-EDGE5-SYM-00-BRST-COHOMOLOGY-v5-02, CONTRIB-EDGE5-SYM-00-BRST-COHOMOLOGY-v5-03. **Rival disposition.** Treating the action as either a representative redundancy or a primitive carrier without a quotient/removal proof. The candidate-kind, physical-disposition, and role predicates separate physical action from skin and carrier architecture.

#### Exact nonclaims

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 7.5 Benchmark 5: Poincaré transformation

**Family:** FAM-05-POINCARE. **Partition certificate:** PART5-FAM-05-POINCARE.

The Poincaré family contains domain anchor, active state action, passive redescription, and Noether witness.

Children are disjoint by level and candidate kind.

**Principal sources:**

citeWigner1939;

citeMaleyRoleOccupancy.

### 7.5.1 Passive Lorentz-coordinate redescription

**Record:** SYM-00-LORENTZ-PASSIVE-v5 **Target:** D-SM-CORE **Constructor:** CTR-PROPER-REDESCRIPTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; Representative/FieldSpace.

**Object and law:** Invertible coordinate/frame change preserving the same geometric fields.  $x' = \Lambda x$  with tensor/spinor components transformed to represent the same geometric object

**Acted upon:** Coordinate components and field representatives.

**Invariant or preserved content:** Geometric observables, state, and physical event.

**Physical projection:** Coordinate representations of one geometry are identified.

**Generated disposition:** TrivialAction.

**Generated status.** Classical Covariant; quantum Exact; anomaly None; vacuum Not Applicable; scale AllTargetScales (Admissible coordinate/frame atlas). **Operator and global scope.** Operator loci: Fields, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** Skin [Representative/FieldSpace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 9 admitted facts, 9 exact source claims, and 9 positive/negative role tests. Selected source basis [20, 25].

- **Certificate 1.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.

- 5 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 1 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-LORENTZ-PASSIVE-v5-01**. **Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.5.2 Active Poincaré action on physical states

**Record:** SYM-00-POINCARÉ-ACTIVE-v5 **Target:** D-SM-CORE **Constructor:** CTR-PHYSICAL-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Unitary representation  $U(a, \Lambda)$  of the connected Poincaré group.  $U(a, \Lambda) O(x) U^{-1} = D(\Lambda) O(\Lambda x + a)$

**Acted upon:** Physical states and local observables.

**Invariant or preserved content:** S-matrix and correlation structure in the flat target.

**Physical projection:** No gauge identification removes active state transformations.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Flat-background relativistic target). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]. **Disposition and resolution.** Normal Form; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 9 admitted facts, 9 exact source claims, and 9 positive/negative role tests. Selected source basis [25].

- **Certificate 1.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- 5 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-POINCARÉ-ACTIVE-v5-01, CONTRIB-EDGE5-SYM-00-POINCARÉ-ACTIVE-v5-02. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.5.3 Poincaré structure as domain anchor

**Record:** SYM-00-POINCARÉ-ANCHOR-v5 **Target:** D-SM-CORE **Constructor:** CTR-DOMAIN-ANCHOR

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Domain/Target.

**Object and law:** Minkowski metric and connected Poincaré group structure. The target admits the isometry group  $ISO(1,3)^{\uparrow}$

**Acted upon:** Definition of the flat relativistic target.

**Invariant or preserved content:** Relativistic domain identity and causal structure.

**Physical projection:** Domain datum, not a representative action.

**Generated disposition:** StructuralDatum.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Flat-background target). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** Anchor [TargetTyping/Target]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 9 admitted facts, 9 exact source claims, and 9 positive/negative role tests. Selected source basis [20, 25].

- **Certificate 1.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- 5 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-POINCARÉ-ANCHOR-v5-01, CONTRIB-EDGE5-SYM-00-POINCARÉ-ANCHOR-v5-02. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.5.4 Poincaré Noether-current diagnostics

**Record:** SYM-00-POINCARÉ-NOETHER-WITNESS-v5 **Target:** D-SM-CORE **Constructor:** CTR-NOETHER-DIAGNOSTIC

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/LocalOperators.

**Object and law:** Energy-momentum and angular-momentum current relations. Translation and Lorentz invariance imply divergence relations under stated boundary assumptions

**Acted upon:** Local operator algebra and conserved charges.

**Invariant or preserved content:** Conserved generators and Ward identities.

**Physical projection:** A diagnostic relation is not an action to quotient.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Boundary assumptions required for integrated charges). **Operator and global scope.** Operator loci: LocalOperators, States. Boundary and global dependence: boundary-sensitive; insensitive within the target to topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** DerivedWitness [Consequence/LocalOperators]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 9 admitted facts, 9 exact source claims, and 9 positive/negative role tests. Selected source basis [20, 25].

- **Certificate 1.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** On Unitary Representations of the Inhomogeneous Lorentz Group; pp. 149-204; persistent authority 10.2307/1968551. The exact certificate ID is retained in the source-claim registry.
- 5 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-POINCARÉ-NOETHER-WITNESS-v5-01, CONTRIB-EDGE5-SYM-00-POINCARÉ-NOETHER-WITNESS-v5-02. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

#### Exact nonclaims

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 7.6 Benchmark 6: CPT

**Family:** FAM-06-CPT. **Partition certificate:** PART5-FAM-06-CPT.

CPT is split into theorem consequence and physical action.

Diagnostic and action children are disjoint by candidate kind.

**Principal sources:**

citeGreavesThomas2014;

citeMaleyFixedDomain.

### 7.6.1 CPT physical discrete action

**Record:** SYM-00-CPT-ACTION-v5 **Target:** D-LOCAL-QFT-CPT **Constructor:** CTR-PHYSICAL-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Antiunitary/unitary-composed CPT operator as fixed by convention. CPT maps each local field to its conjugate at -x with the spin/statistics phases fixed by the formulation

**Acted upon:** Fields, states, particles and antiparticles.

**Invariant or preserved content:** Dynamics and S-matrix under the theorem premises.

**Physical projection:** CPT-related states are not gauge-identified.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum Conditional; scale AllTargetScales (Declared CPT premise target). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: CPTPremiseBundle. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]. **Disposition and resolution.** ConditionalPhysics; record completeness Complete; physics resolution Conditional. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [9, 12].

- **Certificate 1.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.

- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-CPT-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-CPT-ACTION-v5-02, CONTRIB-EDGE5-SYM-00-CPT-ACTION-v5-03. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

The conclusion is restricted to the explicitly named controlled target variant.  
The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 7.6.2 CPT theorem consequence

**Record:** SYM-00-CPT-THEOREM-v5 **Target:** D-LOCAL-QFT-CPT **Constructor:** CTR-DERIVE D-THEOREM

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/Target.

**Object and law:** CPT theorem implication from the declared premise bundle. Locality, Lorentz covariance, spectrum/spin-statistics, and Hermiticity imply CPT invariance in the selected formulation

**Acted upon:** Class of local Lorentz-covariant QFTs.

**Invariant or preserved content:** Existence of the CPT operation and equality constraints.

**Physical projection:** The theorem statement is not itself an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum Conditional; scale AllTargetScales (Exactly the declared CPT premise target). **Operator and global scope.** Operator loci: LocalOperators, States. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: CPTPremiseBundle. Target differences: none. **Role residue.** DerivedWitness [Consequence/Target]. **Disposition and resolution.** ConditionalPhysics; record completeness Complete; physics resolution Conditional. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [9, 12].

- **Certificate 1.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** On the CPT Theorem; Secs. 2-4; persistent authority 10.1016/j.shpsb.2013.10.001. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-CPT-THEOREM-v5-01, CONTRIB-EDGE5-SYM-00-CPT-THEOREM-v5-02, CONTRIB-EDGE5-SYM-00-CPT-THEOREM-v5-03. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

#### Exact nonclaims

The conclusion is restricted to the explicitly named controlled target variant.  
The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

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## Benchmark dossiers II: anomalies, flavor, and vacuum

### 8.1 Benchmark 7: Axial $U(1)_A$

**Family:** FAM-07-U1A. **Partition certificate:** PART5-FAM-07-U1A.

Axial  $U(1)$  is split into classical action, ABJ obstruction, and non-invertible massless-defect action.

Children differ by target and operator category.

**Principal sources:**

`citeAdler1969`;

`citeMaleyRoleOccupancy`;

`citeChoiLamShao2022`.

### 8.1.1 ABJ anomaly diagnostic for $U(1)_A$

**Record:** SYM-00-U1A-ABJ-DIAGNOSTIC-v5 **Target:** D-QCD-MASSLESS-NF3 **Constructor:** CTR-DIAGNOSTIC-OBSTRUCTION

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/LocalOperators.

**Object and law:** Anomalous axial Ward identity.  $\partial_\mu J_5^\mu$  equals the gauge topological density with the standard coefficient plus mass terms

**Acted upon:** Renormalized axial current and gauge-field topological density.

**Invariant or preserved content:** Gauge invariance and the quantum measure; axial current conservation fails.

**Physical projection:** The anomaly equation is not an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical NotApplicable; quantum AnomalousCurrent; anomaly ABJ; vacuum StateDependent; scale AllTargetScales (Renormalized quantum massless target).

**Operator and global scope.** Operator loci: LocalOperators, PartitionFunction. Boundary and global dependence: topology-sensitive; insensitive within the target to boundary-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, NonperturbativeTopology. Realization conditions: MasslessLimit. Target differences: none. **Role residue.** DerivedWitness [Consequence/LocalOperators]; Obstruction [ConsistencyOrMatching/ABJ]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved.

**Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [2, 20].

- **Certificate 1.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U1A-ABJ-DIAGNOSTIC-v5-01, CONTRIB-EDGE5-SYM-00-U1A-ABJ-DIAGNOSTIC-v5-02. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

**Exact nonclaims**

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**8.1.2 Classical axial  $U(1)_A$  action**

**Record:** SYM-00-U1A-CLASSICAL-ACTION-v5 **Target:** D-QCD-MASSLESS-NF3 **Constructor:** CTR-LIMIT-ACTION

**Atomic claim and generated disposition**

**Kind and locus:** Action; PhysicalAction/FieldSpace.

**Object and law:** Global axial phase  $\exp(i\alpha\gamma_5)$ .  $q_L$  maps to  $\exp(i\alpha)q_L$  and  $q_R$  maps to  $\exp(-i\alpha)q_R$

**Acted upon:** Massless quark fields and chirality-sensitive operators.

**Invariant or preserved content:** Classical massless QCD action.

**Physical projection:** The axial action is not a gauge/basis identification.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum AnomalousCurrent; anomaly ABJ; vacuum StateDependent; scale ParameterLimit (All selected quark masses set to zero before quantization). **Operator and global scope.** Operator loci: Fields, LocalOperators, States. Boundary and global dependence: topology-sensitive; insensitive within the target to boundary-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, VacuumSelection, MassDeformation. Realization conditions: MasslessLimit. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/FieldSpace]; EffectiveRoute [Regime/ParameterLimit]; Obstruction [ConsistencyOrMatching/ABJ]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 15 admitted facts, 15 exact source claims, and 9 positive/negative role tests. Selected source basis [2, 20].

- **Certificate 1.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Axial-Vector Vertex in Spinor Electrodynamics; pp. 2426-2438; persistent authority 10.1103/PhysRev.177.2426. The exact certificate ID is retained in the source-claim registry.
- 11 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U1A-CLASSICAL-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-U1A-CLASSICAL-ACTION-v5-02.

**Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.1.3 Non-invertible chiral defect action in a fixed massless target

**Record:** SYM-00-U1A-NONINVERTIBLE-ACTION-v5 **Target:** D-QED-MASSLESS-DEFECT  
**Constructor:** CTR-GENERALIZED-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/ExtendedOperators.

**Object and law:** Topological codimension-one non-invertible chiral defect. Defect crossing implements the chiral transformation with non-group fusion prescribed by the cited construction

**Acted upon:** Local operators and 't Hooft line sectors.

**Invariant or preserved content:** Correlation functions with the topological defect and anomaly-compatible selection rules.

**Physical projection:** Defect action is physical and not a representative identification.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum StateDependent; scale ParameterLimit (Specified massless QED/QCD defect target). **Operator and global scope.** Operator loci: ExtendedOperators, LocalOperators. Boundary and global dependence: global-form-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, topology-sensitive, line-spectrum-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: MassDeformation. Realization conditions: MasslessLimit, ExtendedOperatorsAdmitted, NoDynamicalMonopoles. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/ExtendedOperators]; EffectiveRoute [Regime/ParameterLimit]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 14 admitted facts, 14 exact source claims, and 9 positive/negative role tests. Selected source basis [5, 20].

- **Certificate 1.** Non-Invertible Global Symmetries in the Standard Model; main construction and supplemental derivation; persistent authority 10.1103/PhysRevLett.129.161601. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Non-Invertible Global Symmetries in the Standard Model; main construction and supplemental derivation; persistent authority 10.1103/PhysRevLett.129.161601. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Non-Invertible Global Symmetries in the Standard Model; main construction and supplemental derivation; persistent authority 10.1103/PhysRevLett.129.161601. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Non-Invertible Global Symmetries in the Standard Model; main construction and supplemental derivation; persistent authority 10.1103/PhysRevLett.129.161601. The exact certificate ID is retained in the source-claim registry.
- 10 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U1A-NONINVERTIBLE-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-U1A-NONINVERTIBLE-ACTION-v5-02, CONTRIB-EDGE5-SYM-00-U1A-NONINVERTIBLE-ACTION-v5-03

N-v5-02, CONTRIB-EDGE5-SYM-00-U1A-NONINVERTIBLE-ACTION-v5-03. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 8.2 Benchmark 8: Electroweak B+L

**Family:** FAM-08-BPLUSL. **Partition certificate:** PART5-FAM-08-BPLUSL.

B+L is split into perturbative accidental action and nonperturbative anomaly/topology diagnostic.

Children differ by quantum sector and candidate kind.

**Principal sources:**

citeHooft1976;

citeMaleyRoleOccupancy.

### 8.2.1 Electroweak B+L anomaly and topology diagnostic

**Record:** SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5    **Target:** D-SM-EW-NONPERT    **Constructor:** CTR-DIAGNOSTIC-OBSTRUCTION

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/Topology.

**Object and law:** Anomalous B+L Ward identity and electroweak topological transition. Delta(B+L) is fixed by the electroweak topological charge and generation count

**Acted upon:** B+L current, amplitudes, and selection rules.

**Invariant or preserved content:** Gauge consistency while the global current is not exactly conserved.

**Physical projection:** The violation diagnostic is not an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical NotApplicable; quantum NonperturbativelyViolated; anomaly ElectroweakBPlusL; vacuum StateDependent; scale TopologicalSector (Electroweak instanton/sphaleron sector). **Operator and global scope.** Operator loci: States, PartitionFunction, LocalOperators. Boundary and global dependence: topology-sensitive; insensitive within the target to boundary-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, NonperturbativeTopology. Realization conditions: ElectroweakTopologicalSector. Target differences: none. **Role residue.** DerivedWitness [Consequence/Topology]; Obstruction [ConsistencyOrMatching/ElectroweakBPlusL]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [1, 20].

- **Certificate 1.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5-01, CONTRIB-EDGE5-SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5-02. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

**Exact nonclaims**

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**8.2.2 Perturbative accidental B+L action**

**Record:** SYM-00-BPLUSL-PERT-ACTION-v5 **Target:** D-SM-RENORM-PERT **Constructor:** CT R-LIMIT-ACTION

**Atomic claim and generated disposition**

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Global zero-form B+L phase. Each field transforms by  $\exp(i \text{al-pha}(B+L))$

**Acted upon:** Quark and lepton fields, states, and perturbative amplitudes.

**Invariant or preserved content:** Renormalizable  $d \leq 4$  classical action and perturbative selection rules.

**Physical projection:** B+L is not gauge redundancy.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical AccidentalAtOrder; quantum PerturbativelyExact; anomaly ElectroweakBPlusL; vacuum Invariant; scale OperatorDimension (Renormalizable  $d \leq 4$  perturbative target). **Operator and global scope.** Operator loci: Fields, States, LocalOperators. Boundary and global dependence: topology-sensitive; insensitive within the target to boundary-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, NonperturbativeTopology, HigherOperator. Realization conditions: PerturbativeSector. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]; EffectiveRoute [Regime/OperatorDimension]; Obstruction [ConsistencyOrMatching/ElectroweakBPlusL]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 15 admitted facts, 15 exact source claims, and 9 positive/negative role tests. Selected source basis [1, 20].

- **Certificate 1.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Symmetry Breaking through Bell–Jackiw Anomalies; pp. 8-11; persistent authority 10.1103/PhysRevLett.37.8. The exact certificate ID is retained in the source-claim registry.
- 11 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-BPLUSL-PERT-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-BPLUSL-PERT-ACTION-v5-02. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.3 Benchmark 9: $U(3)^5$ flavor limit

**Family:** FAM-09-U35. **Partition certificate:** PART5-FAM-09-U35.

The  $U(3)^5$  benchmark is the exact zero-Yukawa action child.

Single-child calibration record; basis covariance is owned by family 10.

**Principal sources:**

citeJarlskog1985;

citeMaleySM5.

### 8.3.1 Zero-Yukawa $U(3)^5$ flavor-limit action

**Record:** SYM-00-U35-ZERO-YUKAWA-ACTION-v5 **Target:** D-SM-ZERO-YUKAWA **Constructor:** CTR-LIMIT-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/FieldSpace.

**Object and law:** Independent unitary rotations of the five fermion species spaces.  $Q, u, d, L, e$  transform under  $U(3)_Q \times U(3)_u \times U(3)_d \times U(3)_L \times U(3)_e$

**Acted upon:** Three-generation fermion multiplets.

**Invariant or preserved content:** Kinetic and gauge interactions in the zero-Yukawa limit.

**Physical projection:** The limit group acts on flavor states; basis equivalence of a fixed Yukawa target is a separate record.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum Conditional; anomaly Unresolved; vacuum Invariant; scale ParameterLimit (All Yukawa matrices set to zero). **Operator and global scope.** Operator loci: Fields, States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: ExplicitTerm, Anomaly. Realization conditions: ZeroYukawaLimit, ThreeGenerations. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/FieldSpace]; EffectiveRoute [Regime/ParameterLimit]. **Disposition and resolution.** OpenFrontier; record completeness Complete; physics resolution Open. **Evidence admission.** 14 admitted facts, 14 exact source claims, and 9 positive/negative role tests. Selected source basis [11, 17].

- **Certificate 1.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- 10 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U35-ZERO-YUKAWA-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-U35-ZERO-YUKAWA-ACTION-v5-02.

**Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

The record is structurally classified but its unresolved physics is not reported as exact.

## 8.4 Benchmark 10: CKM/PMNS rephasing

**Family:** FAM-10-MIXING. **Partition certificate:** PART5-FAM-10-MIXING.

Mixing is split into rephasing skin, Jarlskog invariant, and neutrino-target update.

Children differ by quotient, candidate kind, and target.

**Principal sources:**

citeJarlskog1985;

citeMaleySM5.

### 8.4.1 Basis-invariant quark-sector CP diagnostic

**Record:** SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5    **Target:** D-SM-QUARK-MIXING    **Constructor:** CTR-NOETHER-DIAGNOSTIC

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/ParameterSpace.

**Object and law:** Weak-basis invariant commutator/determinant relation. The determinant/trace of commutators is proportional to the Jarlskog invariant and mass-difference factors

**Acted upon:** Up- and down-type Yukawa/mass matrices.

**Invariant or preserved content:** Physical CP-odd residue under weak-basis changes.

**Physical projection:** The invariant is defined on the basis quotient; it is not an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical **Exact**; quantum **MatchingConstrained**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Renormalized quark Yukawa target; running preserves basis-invariant meaning). **Operator and global scope.** Operator loci: **Couplings**, **LocalOperators**. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: **NondegenerateQuarkMasses**, **ThreeGenerations**. Target differences: none. **Role residue.** **DerivedWitness** [Consequence/ParameterSpace]. **Disposition and resolution.** **NormalForm**; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [11, 17].

- **Certificate 1.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5-01**, **CONTRIB-EDGE5-SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5-02**. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive

symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

#### Exact nonclaims

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.4.2 Quark/lepton field rephasing freedom

**Record:** SYM-00-MIXING-REPHASING-SKIN-v5 **Target:** D-SM-QUARK-MIXING **Constructor:** CTR-PROPER-REDESCRIPTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; Representative/FieldSpace.

**Object and law:** Unitary field rephasings preserving kinetic terms. Field phases and Yukawa/CKM coordinates transform together

**Acted upon:** Quark fields and matrix representatives.

**Invariant or preserved content:** Mass eigenvalues, mixing angles after quotient, and CP invariants.

**Physical projection:** Weak-basis related matrix representatives are identified.

**Generated disposition:** TrivialAction.

**Generated status.** Classical Covariant; quantum Exact; anomaly None; vacuum Not Applicable; scale AllTargetScales (Fixed quark target and allowed weak-basis group).

**Operator and global scope.** Operator loci: Fields, Couplings. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: NondegenerateQuarkMasses, ThreeGenerations. Target differences: none. **Role residue.** Skin [Representative/FieldSpace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [11, 17].

- **Certificate 1.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority

10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.

- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-MIXING-REPHASING-SKIN-v5-01, CONTRIB-EDGE5-SYM-00-MIXING-REPHASING-SKIN-v5-02.

**Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.4.3 Neutrino-sector change controlling PMNS rephasing

**Record:** SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5 **Target:** D-SM-NEUTRINO-UPDATE  
**Constructor:** CTR-TARGET-UPDATE

#### Atomic claim and generated disposition

**Kind and locus:** Structure; TargetUpdate/Target.

**Object and law:** Change among massless, Dirac, Weinberg-operator, and Majorana neutrino targets. The target packet changes its fields/operators and therefore its physical phase count

**Acted upon:** Field content, lepton-number structure, and allowed rephasing quotient.

**Invariant or preserved content:** Within each target, basis-invariant mixing data are fixed.

**Physical projection:** Different neutrino targets are not identified by a same-scope quotient.

**Generated disposition:** TargetUpdateDisposition.

**Generated status.** Classical NotApplicable; quantum Conditional; anomaly Unresolved; vacuum Invariant; scale OperatorDimension (Variant-dependent: d=4 Dirac or d=5 Weinberg/Majorana). **Operator and global scope.** Operator loci: Fields, Couplings, States. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: EFTOrder, FieldContent, NeutrinoSector. **Role residue.** TargetUpdate [TargetComparison/Target]. **Disposition and resolution.** Target Update; record completeness Complete; physics resolution NotApplicable. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [11, 17].

- **Certificate 1.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation; pp. 1039-1042; persistent authority 10.1103/PhysRevLett.55.1039. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5-01, CONTRIB-EDGE5-SYM-00-PMNS-NEUTRINO-TARGET-UPD

**ATE-v5-02. Rival disposition.** Treating the changed target as a same-scope counterexample or internal role. The target-difference certificate routes it to TargetUpdate before same-scope role evaluation.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The changed target is not a same-scope rival to the source target.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

The record is structurally classified but its unresolved physics is not reported as exact.

## 8.5 Benchmark 11: Custodial symmetry

**Family:** FAM-11-CUSTODIAL. **Partition certificate:** PART5-FAM-11-CUSTODIAL.

Custodial material is split into the limit action and its diagnostic.

Children differ by action versus diagnostic locus.

**Principal sources:**

citeElitzur1975;

citeMaleyRoleOccupancy;

citeMaleyStandaloneSM.

### 8.5.1 Custodial mass and rho-parameter diagnostic

**Record:** SYM-00-CUSTODIAL-DIAGNOSTIC-v5 **Target:** D-EW-CUSTODIAL-LIMIT **Constructor:** CTR-NOETHER-DIAGNOSTIC

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/LocalOperators.

**Object and law:** Custodial invariant relations among two-point functions and vector masses. In the custodial limit the tree-level rho relation and multiplet degeneracies follow

**Acted upon:** Gauge-invariant electroweak observables.

**Invariant or preserved content:** Custodial-breaking observables and invariant relations.

**Physical projection:** A diagnostic relation is not an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical **ExactInLimit**; quantum **Conditional**; anomaly **Unresolved**; vacuum **Invariant**; scale **ParameterLimit** (Same custodial limit as the action child). **Operator and global scope.** Operator loci: **LocalOperators**, **States**. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: **UnequalCoupling**, **ExplicitTerm**. Realization conditions: **HiggsFixedPhase**. Target differences: none. **Role residue.** **DerivedWitness** [Consequence/LocalOperators]; **EffectiveRoute** [Regime/ParameterLimit]. **Disposition and resolution.** **OpenFrontier**; record completeness **Complete**; physics resolution **Open**. **Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [7, 20].

- **Certificate 1.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-CUSTODIAL-DIAGNOSTIC-v5-01**, **CONTRIB-EDGE5-SYM-00-CUSTODIAL-DIAGNOSTIC-v5-02**. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

**Exact nonclaims**

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

The record is structurally classified but its unresolved physics is not reported as exact.

**8.5.2 Custodial SU(2) physical action in the scalar limit**

**Record:** SYM-00-CUSTODIAL-LIMIT-ACTION-v5 **Target:** D-EW-CUSTODIAL-LIMIT **Constructor:** CTR-LIMIT-ACTION

**Atomic claim and generated disposition**

**Kind and locus:** Action; PhysicalAction/Vacuum.

**Object and law:** Diagonal custodial SU(2)<sub>V</sub> inherited from SU(2)<sub>L</sub> x SU(2)<sub>R</sub>. The Higgs bidoublet transforms as  $\Phi \rightarrow L \Phi R^\dagger$ ; the vacuum preserves the diagonal subgroup

**Acted upon:** Scalar composites, vector multiplets, and vacuum order data.

**Invariant or preserved content:** Scalar potential and vacuum in the custodial limit.

**Physical projection:** The custodial global action is not electroweak gauge redundancy.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum Conditional; anomaly Unresolved; vacuum Invariant; scale ParameterLimit ( $g'=0$  and custodial-breaking Yukawa splittings removed). **Operator and global scope.** Operator loci: LocalOperators, States, Vacuum. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: UnequalCoupling, ExplicitTerm. Realization conditions: HiggsFixedPhase. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/Vacuum]; EffectiveRoute [Regime/ParameterLimit]. **Disposition and resolution.** OpenFrontier; record completeness Complete; physics resolution Open. **Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [7, 20, 22].

- **Certificate 1.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-CUSTODIAL-LIMIT-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-CUSTODIAL-LIMIT-ACTION-v5-02, CONTRIB-EDGE5-SYM-00-CUSTODIAL-LIMIT-ACTION-v5-03. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

The record is structurally classified but its unresolved physics is not reported as exact.

## 8.6 Benchmark 12: Electroweak gauge breaking

**Family:** FAM-12-EW-FIXATION. **Partition certificate:** PART5-FAM-12-EW-FIXATION.

Electroweak fixation is split into proper gauge skin, fixation structure, residual  $U(1)$  action, and Goldstone representative.

Children are disjoint by candidate kind and quotient disposition.

**Principal sources:**

citeHenneauxTeitelboim1992;

citeMaleySM2;

citeElitzur1975;

citeMaleyRoleOccupancy;

citeMaleyStandaloneSM.

### 8.6.1 Local electroweak gauge operation in the Higgs phase

**Record:** SYM-00-EW-LOCAL-GAUGE-SKIN-v5 **Target:** D-EW-HIGGS-PHASE **Constructor:** CTR-PROPER-REDESCRIPTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; Representative/Bulk.

**Object and law:** Proper  $SU(2)_L \times U(1)_Y$  gauge transformation. Standard electroweak gauge transformation of W, B, and Higgs representatives

**Acted upon:** Gauge-fixed Higgs and gauge-field representatives.

**Invariant or preserved content:** Gauge-invariant phase and observable spectrum.

**Physical projection:** Proper local gauge-related configurations are identified.

**Generated disposition:** TrivialAction.

**Generated status.** Classical Exact; quantum Exact; anomaly None; vacuum Higgs Fixed; scale AllTargetScales (Selected one-doublet electroweak target). **Operator and global scope.** Operator loci: Fields, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: HiggsFixedPhase. Target differences: none. **Role residue.** Skin [Representative/Bulk]. **Disposition and resolution.** Normal Form; record completeness Complete; physics resolution Resolved. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [10, 18].

- **Certificate 1.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Quantization of Gauge Systems; Chs. 1, 3, 13; persistent authority bundled frozen snapshot. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-EW-LOCAL-GAUGE-SKIN-v5-01, CONTRIB-EDGE5-SYM-00-EW-LOCAL-GAUGE-SKIN-v5-02.

**Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.6.2 Electroweak vacuum/fixation structure

**Record:** SYM-00-EW-VACUUM-FIXATION-v5 **Target:** D-EW-HIGGS-PHASE **Constructor:** CTR-BRIDGE-FIXATION

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Vacuum/Vacuum.

**Object and law:** Gauge-invariant Higgs bridge and vacuum stabilizer data. The selected Higgs-phase configuration fixes mass-response directions and a residual electromagnetic stabilizer

**Acted upon:** Electroweak response architecture and phase.

**Invariant or preserved content:** Gauge-invariant spectrum, charges, and phase diagnostics.

**Physical projection:** Fixation is structural phase data, not a local gauge operation.

**Generated disposition:** StructuralDatum.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum Higgs **Fixed**; scale **AllTargetScales** (Selected electroweak phase). **Operator and global scope.** Operator loci: Vacuum, LocalOperators, States. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: HiggsFixedPhase. Target differences: none. **Role residue.** CarrierConstraint [Architecture/Vacuum]; BridgeFixation [Transition/Vacuum]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [7, 20].

- **Certificate 1.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-EW-VACUUM-FIXATION-v5-01, CONTRIB-EDGE5-SYM-00-EW-VACUUM-FIXATION-v5-02, CONTRIB-EDGE5-SYM-00-EW-VACUUM-FIXATION-v5-03. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

**Exact nonclaims**

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**8.6.3 Gauge-fixed Goldstone representative**

**Record:** SYM-00-GOLDSTONE-REPRESENTATIVE-v5 **Target:** D-EW-HIGGS-PHASE **Constructor:** CTR-PROPER-REDESCRIPTION

**Atomic claim and generated disposition**

**Kind and locus:** Action; Representative/FieldSpace.

**Object and law:** Gauge choice and field-coordinate change exposing Goldstone components.  $R_{\xi}$ /unitary-gauge field redefinitions and gauge fixing redistribute scalar and longitudinal coordinates

**Acted upon:** Gauge-fixed scalar and longitudinal-vector representatives.

**Invariant or preserved content:** Gauge-invariant poles and S-matrix.

**Physical projection:** Gauge-related field coordinatizations are identified.

**Generated disposition:** TrivialAction.

**Generated status.** Classical Covariant; quantum Exact; anomaly None; vacuum Higgs Fixed; scale AllTargetScales (Perturbative gauges within the fixed Higgs phase). **Operator and global scope.** Operator loci: Fields, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: HiggsFixedPhase, GaugeFixingChosen. Target differences: none. **Role residue.** Skin [Representative/FieldSpace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [7, 20].

- **Certificate 1.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GOLDSTONE-REPRESENTATIVE-v5-01, CONTRIB-EDGE5-SYM-00-GOLDSTONE-REPRESENTATIVE-v5-02. **Rival disposition.** Treating a quotient-trivial representative change as a physical action. The representative-equivalence fact shows invariant quotient content is unchanged.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 8.6.4 Residual electromagnetic physical action

**Record:** SYM-00-U1EM-RESIDUAL-ACTION-v5 **Target:** D-EW-HIGGS-PHASE **Constructor:** CTR-PHYSICAL-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Global electromagnetic  $U(1)_{\text{em}}$  charge action. A state of electric charge  $Q$  transforms by  $\exp(i \alpha Q)$

**Acted upon:** Physical charged states and gauge-invariant charged operators.

**Invariant or preserved content:** Electroweak Higgs-phase dynamics and charge supers-election.

**Physical projection:** The physical charge action survives the proper-gauge quotient.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **AllTargetScales** (Selected electroweak Higgs phase). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: boundary-sensitive; insensitive within the target to topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: HiggsFixedPhase. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [7, 22].

- **Certificate 1.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Impossibility of Spontaneously Breaking Local Symmetries; pp. 3978-3982; persistent authority 10.1103/PhysRevD.12.3978. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-U1EM-RESIDUAL-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-U1EM-RESIDUAL-ACTION-v5-02, CONTRIB-EDGE5-SYM-00-U1EM-RESIDUAL-ACTION-v5-03. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

**Exact nonclaims**

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

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## Benchmark dossiers III: global form, generalized structure, scale, and effective limits

### 9.1 Benchmark 13: $G_{SM}/\Gamma$

**Family:** FAM-13-GLOBAL-FORM. **Partition certificate:** PART5-FAM-13-GLOBAL-FORM.

Global form is split into target update, selected Z6 anchor, and derived line-spectrum structure.

Old-target update and updated-target anchor are explicitly indexed by different targets.

**Principal sources:**

`citeAharonySeibergTachikawa2013;`

`citeMaleyRoleOccupancy.`

### 9.1.1 Faithful Standard Model global-form choice

**Record:** SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5 **Target:** D-SM-GLOBAL-FORM-UPDATE  
**Constructor:** CTR-TARGET-UPDATE

#### Atomic claim and generated disposition

**Kind and locus:** Structure; TargetUpdate/Target.

**Object and law:** Selection of Gamma in  $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\text{Gamma}$ . The quotient subgroup identifies center elements and changes allowed global sectors

**Acted upon:** Faithful gauge group, bundles, line spectrum, monopoles, and theta periodicities.

**Invariant or preserved content:** Local Lie algebra and perturbative field representations can remain fixed while global theory data change.

**Physical projection:** Different faithful global forms are distinct controlled targets.

**Generated disposition:** TargetUpdateDisposition.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State Dependent**; scale **AllTargetScales** (Global-form family of targets). **Operator and global scope.** Operator loci: ExtendedOperators, PartitionFunction. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: GlobalForm, OperatorCategory, Topology. **Role residue.** TargetUpdate [TargetComparison/Target]. **Disposition and resolution.** TargetUpdate; record completeness **Complete**; physics resolution **NotApplicable**. **Evidence admission.** 11 admitted facts, 11 exact source claims, and 9 positive/negative role tests. Selected source basis [3, 20].

- **Certificate 1.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- 7 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5-01, CONTRIB-EDGE5-SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5-02. **Rival disposition.** Treating the changed target as a same-scope counterexample or internal role. The target-difference certificate routes it to TargetUpdate before same-scope role evaluation.

**Exact nonclaims**

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The changed target is not a same-scope rival to the source target.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**9.1.2 Selected Z6 faithful Standard Model gauge group**

**Record:** SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5   **Target:** D-SM-DEFECT-Z6   **Constructor:** CTR-DOMAIN-ANCHOR

**Atomic claim and generated disposition**

**Kind and locus:** Structure; TargetData/Target.

**Object and law:** Faithful gauge group  $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\text{Z}_6$ . The quotient fixes the faithful representations and admissible extended-operator lattice.

**Acted upon:** Bundles, genuine lines, monopole sectors, and theta identifications.

**Invariant or preserved content:** Faithful group, bundle classes, and genuine line spectrum.

**Physical projection:** Not an operation; the global form is target data..

**Generated disposition:** StructuralDatum.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Higgs Fixed**; scale **AllTargetScales** (Global-form family of targets). **Operator and global scope.** Operator loci: **ExtendedOperators**, **PartitionFunction**. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive; insensitive within the target to boundary-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: **FixedGlobalFormZ6**, **NoDynamicalMonopoles**, **ExtendedOperatorsAdmitted**. Target differences: none. **Role residue.** Anchor [TargetTyping/Target]. **Disposition and resolution.** **NormalForm**; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [3, 20].

- **Certificate 1.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: `CONTRIB-EDGE5-SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5-04`, `CONTRIB-EDGE5-SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5-05`. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.1.3 Genuine line-operator spectrum for a fixed global form

**Record:** `SYM-00-LINE-SPECTRUM-STRUCTURE-v5` **Target:** `D-SM-DEFECT-Z6` **Constructor:** `CTR-STRUCTURAL-WITNESS`

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Architecture/ExtendedOperators.

**Object and law:** Charge lattice, mutual-locality relation, and screening equivalence of genuine lines. Global form and dynamical matter determine which line charges are genuine and which are screened

**Acted upon:** Definition of admissible Wilson, 't Hooft, and dyonic line sectors.

**Invariant or preserved content:** Extended-operator spectrum of the fixed theory.

**Physical projection:** The line spectrum is structural data, not an acting symmetry.

**Generated disposition:** `StructuralDatum`.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State Dependent**; scale **AllTargetScales** (Fixed Gamma, matter screening, and monopole content).

**Operator and global scope.** Operator loci: `ExtendedOperators`. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: `FixedGlobalFormZ6`, `NoDynamicalMonopoles`, `ExtendedOperatorsAdmitted`. Target differences: none. **Role residue.** `DerivedWitness` [`Consequence/ExtendedOperators`]. **Disposition and resolution.** `NormalForm`; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [3, 20].

- **Certificate 1.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-LINE-SPECTRUM-STRUCTURE-v5-01, CONTRIB-EDGE5-SYM-00-LINE-SPECTRUM-STRUCTURE-v5-02, CONTRIB-EDGE5-SYM-00-LINE-SPECTRUM-STRUCTURE-v5-03. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 9.2 Benchmark 14: One-form/higher-group flavor structure

**Family:** FAM-14-GENERALIZED. **Partition certificate:** PART5-FAM-14-GENERALIZED.

Generalized symmetry is split into one-form action, higher-group extension datum, and background action.

Children differ by acting object and structure/action kind.

**Principal sources:**

citeCordovaKoren2023;

citeMaleyRoleOccupancy;

citeAharonySeibergTachikawa2013.

### 9.2.1 Combined higher-flavor background action

**Record:** SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 **Target:** D-SM-HIGHER-FLAVOR-Z  
6 **Constructor:** CTR-GENERALIZED-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/ExtendedOperators.

**Object and law:** Coupled zero-form flavor and one-form magnetic-hypercharge background transformation. Zero-form and one-form gauge transformations act jointly according to the fixed extension class

**Acted upon:** Background fields, charged local operators, and genuine lines.

**Invariant or preserved content:** Background-coupled partition function up to the declared 't Hooft anomaly.

**Physical projection:** The combined background transformation probes physical operator/defect charges.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum tHooftAnomalous; anomaly t Hooft; vacuum StateDependent; scale ParameterLimit (Same fixed higher-flavor target and background category). **Operator and global scope.** Operator loci: LocalOperators, ExtendedOperators, PartitionFunction. Boundary and global dependence: global-form-sensitive, line-spectrum-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, topology-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: MassDeformation. Realization conditions: FixedGlobalFormZ6, NoDynamicalMonopoles, ExtendedOperatorsAdmitted, BackgroundFieldsAdmitted, ZeroYukawaLimit. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/ExtendedOperators]; EffectiveRoute [Regime/ParameterLimit]; Obstruction [ConsistencyOrMatching/tHooft]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 17 admitted facts, 17 exact source claims, and 9 positive/negative role tests. Selected source basis [6, 20].

- **Certificate 1.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- 13 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5-02, CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5-03.

**Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.2.2 Higher-flavor extension class

**Record:** SYM-00-HIGHER-FLAVOR-EXTENSION-v5 **Target:** D-SM-HIGHER-FLAVOR-Z6 **Constructor:** CTR-STRUCTURAL-WITNESS

#### Atomic claim and generated disposition

**Kind and locus:** Structure; Architecture/Target.

**Object and law:** Nontrivial extension of zero-form flavor backgrounds by magnetic hypercharge one-form background data. Background gauge transformations obey the extension/2-group compatibility specified by the mixed anomaly data

**Acted upon:** Background-field consistency and combined symmetry structure.

**Invariant or preserved content:** Combined partition function and background consistency.

**Physical projection:** The extension class is structural data, not itself an action.

**Generated disposition:** StructuralDatum.

**Generated status.** Classical `ExactInLimit`; quantum `tHooftAnomalous`; anomaly `tHooft`; vacuum `StateDependent`; scale `ParameterLimit` (Massless flavor target with magnetic hypercharge background and no dynamical monopoles). **Operator and global scope.** Operator loci: `ExtendedOperators`, `PartitionFunction`. Boundary and global dependence: global-form-sensitive, line-spectrum-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, topology-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: `MassDeformation`. Realization conditions: `FixedGlobalFormZ6`, `NoDynamicalMonopoles`, `ExtendedOperatorsAdmitted`, `BackgroundFieldsAdmitted`, `ZeroYukawaLimit`. Target differences: none. **Role residue.** `DerivedWitness` [`Consequence/Target`]; `EffectiveRoute` [`Regime/ParameterLimit`]; `Obstruction` [`ConsistencyOrMatching/tHooft`]. **Disposition and resolution.** `NormalForm`; record completeness `Complete`; physics resolution `Resolved`. **Evidence admission.** 17 admitted facts, 17 exact source claims, and 9 positive/negative role tests. Selected source basis [6, 20].

- **Certificate 1.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.

- **Certificate 3.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Higher Flavor Symmetries in the Standard Model; Secs. 2-5; persistent authority 10.1002/andp.202300031. The exact certificate ID is retained in the source-claim registry.
- 13 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 3 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-EXTENSION-v5-01, CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-EXTENSION-v5-02, CONTRIB-EDGE5-SYM-00-HIGHER-FLAVOR-EXTENSION-v5-03. **Rival disposition.** Treating target/architectural data as an ordinary group action on states. The structure branch is non-action typed; target and removal certificates determine anchor/carrier/update status.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.2.3 Magnetic one-form action in a fixed defect-completed target

**Record:** SYM-00-ONEFORM-ACTION-v5 **Target:** D-SM-DEFECT-Z6 **Constructor:** CTR-GENE  
RALIZED-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/ExtendedOperators.

**Object and law:** Topological codimension-two one-form symmetry operator. Linking the topological surface with a line multiplies the line by its one-form charge phase

**Acted upon:** Genuine line operators.

**Invariant or preserved content:** Correlation functions up to the prescribed charge action.

**Physical projection:** The topological operator acts on genuine lines and is not a gauge representative.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **State** **Dependent**; scale **AllTargetScales** (Fixed global form, screening data, and absence/presence of monopoles as specified). **Operator and global scope.** Operator loci: **ExtendedOperators**. Boundary and global dependence: topology-sensitive, global-form-sensitive, line-spectrum-sensitive, defect-sensitive; insensitive within the target to boundary-sensitive, theta-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: **FixedGlobalFormZ6**, **NoDynamicalMonopoles**, **ExtendedOperatorsAdmitted**. Target differences: none. **Role residue.** **PhysicalAction** [**PhysicalAction/ExtendedOperators**]. **Disposition and resolution.** **NormalForm**; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [3].

- **Certificate 1.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Reading between the Lines of Four-Dimensional Gauge Theories; Secs. 1-3; persistent authority 10.1007/JHEP08(2013)115. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-ONEFORM-ACTION-v5-01**, **CONTRIB-EDGE5-SYM-00-ONEFORM-ACTION-v5-02**. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

**Exact nonclaims**

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.3 Benchmark 15: Classical scale symmetry and trace anomaly

**Family:** FAM-15-SCALE-RG. **Partition certificate:** PART5-FAM-15-SCALE-RG.

Scale/RG material is split into classical action, trace diagnostic, scheme/scale redescrptions, physical flow, matching, and fixed-point rule.

Children are disjoint by candidate kind, target, and disposition.

**Principal sources:**

citeCallanColemanJackiw1970;

citeMaleySM4.

### 9.3.1 EFT threshold-matching transport

**Record:** SYM-00-EFT-THRESHOLD-MATCHING-v5 **Target:** D-SM-EFT-MATCHING-RULE **Constructor:** CTR-THEORYSPACE-TRANSPORT

#### Atomic claim and generated disposition

**Kind and locus:** Transport; TheorySpace/TheorySpace.

**Object and law:** Matching map between specified UV and IR operator bases. IR coefficients are fixed so matched amplitudes agree to the declared EFT order

**Acted upon:** Wilson coefficients, operators, and amplitudes.

**Invariant or preserved content:** Low-energy observables through the truncation order.

**Physical projection:** Matching is a theory-space transport, not a state-space symmetry.

**Generated disposition:** TheorySpaceTransport.

**Generated status.** Classical NotApplicable; quantum MatchingConstrained; anomaly None; vacuum StateDependent; scale EFTOrder (Specified operator dimension and matching scale). **Operator and global scope.** Operator loci: Couplings, TheorySpace, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: HigherOperator. Realization conditions: EFTMatchingPairFixed. Target differences: none. **Role residue.** EffectiveRoute [Regime/EFTOrder]. **Disposition and resolution.** ConditionalRule; record completeness Complete; physics resolution Conditional. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-EFT-THRESHOLD-MATCHING-v5-01, CONTRIB-EDGE5-SYM-00-EFT-THRESHOLD-MATCHING-v5-02. **Rival disposition.** Treating scheme/scale transport as an internal global symmetry on a fixed Hilbert space. The theory-space disposition and prediction-preservation test distinguish redescription from physical flow/matching.

**Exact nonclaims**

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**9.3.2 Exact scale action at a specified fixed point**

**Record:** SYM-00-FIXED-POINT-SCALE-ACTION-v5 **Target:** D-CFT-FIXED-POINT-RULE

**Constructor:** CTR-FIXEDPOINT-RULE

**Atomic claim and generated disposition**

**Kind and locus:** Action; PhysicalAction/LocalOperators.

**Object and law:** Dilatation generator at  $\beta_i=0$ . Operators transform with fixed scaling dimensions and correlators obey fixed-point scaling laws

**Acted upon:** Fixed-point states, local operators, and correlators.

**Invariant or preserved content:** Fixed-point action and conformal/scale correlators under the stated assumptions.

**Physical projection:** The fixed-point action is physical, not a scheme identification.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Exact**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **FixedPoint** (All beta functions vanish at the specified fixed point). **Operator and global scope.** Operator loci: LocalOperators, States. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: FixedPoint. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/LocalOperators]; EffectiveRoute [Regime/FixedPoint]. **Disposition and resolution.** ConditionalRule; record completeness **Complete**; physics resolution **Conditional**. **Evidence admission.** 12 admitted facts, 12 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 8 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-FIXED-POINT-SCALE-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-FIXED-POINT-SCALE-ACTION-v5-02. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.3.3 Physical renormalization-group flow data

**Record:** SYM-00-RG-PHYSICAL-FLOW-v5 **Target:** D-SM-RG-FLOW **Constructor:** CTR-THEORYSPACE-TRANSPORT

#### Atomic claim and generated disposition

**Kind and locus:** Transport; TheorySpace/TheorySpace.

**Object and law:** RG vector field  $\beta_i(g)$  and anomalous-dimension matrix.  $\mu dg_i/d\mu = \beta_i$  and  $\mu dO/d\mu = \gamma O$

**Acted upon:** Theory-space points, couplings, and operator basis.

**Invariant or preserved content:** Consistently transported physical amplitudes.

**Physical projection:** Flow relates scale-indexed theory descriptions; it is not an action quotient on a fixed Hilbert space.

**Generated disposition:** TheorySpaceTransport.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum NotApplicable; scale MatchingInterval (Frozen RG interval and threshold content).

**Operator and global scope.** Operator loci: Couplings, TheorySpace, LocalOperators.

Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive,

theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: RGDeparture. Realization conditions: none. Target differences: none. **Role residue.** EffectiveRoute

[Regime/MatchingInterval]. **Disposition and resolution.** NormalForm; record completeness

Complete; physics resolution Resolved. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 1 retained edge(s) with removal-audited contribution IDs: **CONTRIB-EDGE5-SYM-00-RG-PHYSICAL-FLOW-v5-01**. **Rival disposition.** Treating scheme/scale transport as an internal global symmetry on a fixed Hilbert space. The theory-space disposition and prediction-preservation test distinguish redescription from physical flow/matching.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.3.4 Renormalization-scale coordinate choice

**Record:** SYM-00-RG-SCALE-COORDINATE-v5 **Target:** D-SM-RG-SCHEME **Constructor:** CTR-THEORYSPACE-REDESCRIPTION

#### Atomic claim and generated disposition

**Kind and locus:** Transport; Representative/TheorySpace.

**Object and law:** Choice of reference renormalization scale with compensating running. Changing  $\mu$  is compensated by RG evolution so that a fixed physical prediction is unchanged

**Acted upon:** Coupling/operator coordinates.

**Invariant or preserved content:** RG-invariant observables.

**Physical projection:** Reference-scale choices on the same RG solution are representative coordinates.

**Generated disposition:** TheorySpaceRedescription.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum NotApplicable; scale MatchingInterval (Interval without an unaccounted threshold).

**Operator and global scope.** Operator loci: Couplings, TheorySpace. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** Skin [Representative/TheorySpace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved.

**Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 1 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-RG-SCALE-COORDINATE-v5-01. **Rival disposition.** Treating scheme/scale transport as an internal global symmetry on a fixed Hilbert space. The theory-space disposition and prediction-preservation test distinguish redescription from physical flow/matching.

**Exact nonclaims**

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**9.3.5 Renormalization-scheme reparameterization**

**Record:** SYM-00-RG-SCHEME-REDESCRIPTION-v5 **Target:** D-SM-RG-SCHEME **Constructor:** CTR-THEORYSPACE-REDESCRIPTION

**Atomic claim and generated disposition**

**Kind and locus:** Transport; Representative/TheorySpace.

**Object and law:** Invertible analytic reparameterization of couplings and operator basis.  $g' = f(g)$  with transformed beta functions and Wilson coefficients preserving observables

**Acted upon:** Theory-space coordinates and renormalized operator representatives.

**Invariant or preserved content:** Matched physical amplitudes and predictions.

**Physical projection:** Analytically related scheme coordinates representing the same physics are identified.

**Generated disposition:** TheorySpaceRedescription.

**Generated status.** Classical NotApplicable; quantum Exact; anomaly None; vacuum NotApplicable; scale AllTargetScales (Overlap domain of the two schemes). **Operator and global scope.** Operator loci: Couplings, TheorySpace. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: none. Realization conditions: none. Target differences: none. **Role residue.** Skin [Representative/TheorySpace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 10 admitted facts, 10 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 6 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 1 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-RG-SCHEME-REDESCRIPTION-v5-01. **Rival disposition.** Treating scheme/scale transport as an internal global symmetry on a fixed Hilbert space. The theory-space disposition and prediction-preservation test distinguish redescription from physical flow/matching.

#### Exact nonclaims

Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.3.6 Classical scale action in a massless target

**Record:** SYM-00-SCALE-CLASSICAL-ACTION-v5 **Target:** D-SM-MASSLESS-SCALE **Constructor:** CTR-LIMIT-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/FieldSpace.

**Object and law:** Dilatation of coordinates and fields.  $x \rightarrow \lambda x$  and  $\Phi(x) \rightarrow \lambda^{-\Delta} \Phi(\lambda^{-1}x)$

**Acted upon:** Classical fields and local operators.

**Invariant or preserved content:** Classical action in the dimensionless/massless limit.

**Physical projection:** Dilatations are not representative identifications.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum AnomalousCurrent; anomaly Trace; vacuum StateDependent; scale ParameterLimit (All explicit dimensionful deformations removed). **Operator and global scope.** Operator loci: Fields, LocalOperators, States. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, MassDeformation, RGDeparture. Realization conditions: MasslessLimit. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/FieldSpace]; EffectiveRoute [Regime/ParameterLimit]; Obstruction [ConsistencyOrMatching/Trace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 15 admitted facts, 15 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.

- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 11 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-SCALE-CLASSICAL-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-SCALE-CLASSICAL-ACTION-v5-02.

**Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

### 9.3.7 Quantum trace-anomaly diagnostic

**Record:** SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5 **Target:** D-SM-RENORMALIZED-QFT  
**Constructor:** CTR-DIAGNOSTIC-OBSTRUCTION

#### Atomic claim and generated disposition

**Kind and locus:** Diagnostic; Diagnostic/LocalOperators.

**Object and law:** Renormalized trace Ward identity.  $T^{\mu}_{\mu}$  equals  $\beta_i O_i$  plus mass and anomalous-dimension terms modulo operator mixing

**Acted upon:** Improved energy-momentum tensor, operators, and correlators.

**Invariant or preserved content:** Scheme-independent physical scaling violation and dimensional transmutation.

**Physical projection:** The trace relation is a diagnostic, not an action.

**Generated disposition:** DiagnosticNotAction.

**Generated status.** Classical NotApplicable; quantum AnomalousCurrent; anomaly Trace; vacuum StateDependent; scale AllTargetScales (Renormalized quantum target with fixed operator basis). **Operator and global scope.** Operator loci: LocalOperators, PartitionFunction. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: Anomaly, MassDeformation, RGDeparture. Realization conditions: none. Target differences: none. **Role residue.** DerivedWitness [Consequence/LocalOperators]; Obstruction [ConsistencyOrMatching/Trace]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [4, 16].

- **Certificate 1.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** A New Improved Energy-Momentum Tensor; Secs. 2-4; persistent authority 10.1016/0003-4916(70)90394-5. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5-01, CONTRIB-EDGE5-SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5-02. **Rival disposition.** Treating a current, theorem, anomaly, or invariant as the primitive symmetry or as an ordinary state action. Dependency and obstruction certificates type the object as witness/rule/obstruction rather than action.

**Exact nonclaims**

An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## 9.4 Benchmark 16: Isospin or heavy-quark symmetry

**Family:** FAM-16-EFFECTIVE. **Partition certificate:** PART5-FAM-16-EFFECTIVE.

Effective symmetry is represented by separate isospin and HQET action children.

Children occupy distinct EFT targets and expansion parameters.

**Principal sources:**

citeNeubert1994;

citeMaleySM4;

citeMaleyRoleOccupancy;

citeGasserLeutwyler1984;

citeMaleyRPI.

### 9.4.1 Heavy-quark spin/ flavor action at leading HQET order

**Record:** SYM-00-HQET-SPIN-FLAVOR-ACTION-v5 **Target:** D-HQET **Constructor:** CTR-LIMIT-ACTION

#### Atomic claim and generated disposition

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Heavy-quark spin rotations and heavy-flavor rotations at leading order. The leading HQET Lagrangian is independent of heavy spin and mass after velocity-field redefinition

**Acted upon:** HQET fields and heavy-hadron states.

**Invariant or preserved content:** Leading HQET action and heavy-hadron relations.

**Physical projection:** The action transforms effective physical states.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical ExactInLimit; quantum MatchingConstrained; anomaly None; vacuum Invariant; scale AsymptoticExpansion (Leading order in  $1/m_Q$  with specified QCD matching). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: MassDeformation, HigherOperator. Realization conditions: HeavyQuarkLimit, HadronicRegime. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]; EffectiveRoute [Regime/AsymptoticExpansion]. **Disposition and resolution.** NormalForm; record completeness Complete; physics resolution Resolved. **Evidence admission.** 14 admitted facts, 14 exact source claims, and 9 positive/negative role tests. Selected source basis [16, 20, 24].

- **Certificate 1.** Heavy-Quark Symmetry; Secs. 2-4; persistent authority 10.1016/0370-1573(94)90091-4. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Heavy-Quark Symmetry; Secs. 2-4; persistent authority 10.1016/0370-1573(94)90091-4. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Heavy-Quark Symmetry; Secs. 2-4; persistent authority 10.1016/0370-1573(94)90091-4. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Heavy-Quark Symmetry; Secs. 2-4; persistent authority 10.1016/0370-1573(94)90091-4. The exact certificate ID is retained in the source-claim registry.
- 10 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-HQET-SPIN-FLAVOR-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-HQET-SPIN-FLAVOR-ACTION-v5-02.

**2. Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

**Exact nonclaims**

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

**9.4.2 Approximate isospin SU(2) action**

**Record:** SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5 **Target:** D-QCD-ISOSPIN **Constructor:** CTR-LIMIT-ACTION

**Atomic claim and generated disposition**

**Kind and locus:** Action; PhysicalAction/StateSpace.

**Object and law:** Vector SU(2) rotation of the u,d doublet.  $(u,d)^T \rightarrow U(u,d)^T$  in the equal-mass, no-electromagnetism limit

**Acted upon:** Quark operators and hadronic states.

**Invariant or preserved content:** QCD action in the exact limit and hadronic multiplet relations.

**Physical projection:** Isospin acts on physical states.

**Generated disposition:** NontrivialAction.

**Generated status.** Classical **Approximate**; quantum **Exact**; anomaly **None**; vacuum **Invariant**; scale **IRRegime** (Hadronic regime with corrections controlled by  $m_u, m_d$  and  $e$ ). **Operator and global scope.** Operator loci: States, LocalOperators. Boundary and global dependence: no registered sensitivity; insensitive within the target to boundary-sensitive, topology-sensitive, global-form-sensitive, line-spectrum-sensitive, theta-sensitive, defect-sensitive. **Causal ancestry.** Failure causes: MassDeformation, UnequalCoupling. Realization conditions: HadronicRegime. Target differences: none. **Role residue.** PhysicalAction [PhysicalAction/StateSpace]; EffectiveRoute [Regime/IRRegime]. **Disposition and resolution.** NormalForm; record completeness **Complete**; physics resolution **Resolved**. **Evidence admission.** 13 admitted facts, 13 exact source claims, and 9 positive/negative role tests. Selected source basis [8, 20, 21].

- **Certificate 1.** Chiral Perturbation Theory to One Loop; Secs. 2-4; persistent authority 10.1016/0003-4916(84)90242-2. The exact certificate ID is retained in the source-claim registry.
- **Certificate 2.** Chiral Perturbation Theory to One Loop; Secs. 2-4; persistent authority 10.1016/0003-4916(84)90242-2. The exact certificate ID is retained in the source-claim registry.
- **Certificate 3.** Chiral Perturbation Theory to One Loop; Secs. 2-4; persistent authority 10.1016/0003-4916(84)90242-2. The exact certificate ID is retained in the source-claim registry.
- **Certificate 4.** Chiral Perturbation Theory to One Loop; Secs. 2-4; persistent authority 10.1016/0003-4916(84)90242-2. The exact certificate ID is retained in the source-claim registry.
- 9 additional source claim(s) are listed in the machine-readable evidence registry.

**Support.** 2 retained edge(s) with removal-audited contribution IDs: CONTRIB-EDGE5-SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5-01, CONTRIB-EDGE5-SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5-02. **Rival disposition.** Treating the quotient-nontrivial action as mere skin. The explicit action witness changes states, charges, selection rules, or defects after quotient.

#### Exact nonclaims

A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.

The conclusion is restricted to the explicitly named controlled target variant.

The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.

## Collected Theorem Package

### 10.1 Dependency chain

$$\begin{aligned}
T0 &\Rightarrow T1 \Rightarrow T2 \Rightarrow (T3, T4, T5, T6), \\
&\Rightarrow (T7, T8, T9) \Rightarrow T10, \\
&\Rightarrow (T11, T12, T13, T14, T15) \Rightarrow T16.
\end{aligned}$$

The chain is asymmetric. Kernel necessity and fixed-domain standing are upstream. Physical source theorems specialize the fixed domain. Independent role positions and residues then define a finite occupation problem. The local singleton theorem is proved before apex uniqueness is imported to close residual selector or regime plurality.

Table 10.1: SYM-0 theorem inventory.

| ID      | Name                               | Statement                                                                                                                                                      |
|---------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SYM0-T0 | Kernel necessity                   | Every determinate same-domain symmetry claim instantiates Admissibility, Standing, Reference, and Irreversibility.                                             |
| SYM0-T1 | No neutral exterior                | Kernel denial either preserves the kernel functions extensionally or destroys/changes the determinate target.                                                  |
| SYM0-T2 | Target-update separation           | Changes in target-bearing coordinates are not same-scope rivals or repairs.                                                                                    |
| SYM0-T3 | Registered benchmark decomposition | The declared sixteen-family calibration universe is partitioned into forty-six atomic records; full Standard Model candidate exhaustion is deferred to SYM-11. |
| SYM0-T4 | Kind-indexed disposition           | Actions, structures, diagnostics, and transports enter distinct physical-disposition branches.                                                                 |
| SYM0-T5 | Status-coordinate non-collapse     | No ordinary status coordinate determines the complete benchmark normal form.                                                                                   |
| SYM0-T6 | Causal-ancestry preservation       | Anomaly, deformation, vacuum, fixation, topology, boundary, EFT, and target change remain distinct.                                                            |
| SYM0-T7 | Role-realization completeness      | Every family certificate contains independent role-position, residue, quotient, eligibility, compatibility, failure, and orbit evidence.                       |

| ID       | Name                               | Statement                                                                                                                                                              |
|----------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SYM0-T8  | Eligibility and compatibility      | All role-position/residue pairs are evaluated and every complete preassignment is tested by substantive joint compatibility.                                           |
| SYM0-T9  | Lawful-redescription quotient      | The full invariant-preserving transformation class acts on the assignment problem and preserves the lawful orbit.                                                      |
| SYM0-T10 | Singleton role occupation          | Each role-realization-complete benchmark family has exactly one assignment orbit modulo $\Gamma_{\mathrm{lawful}}$ .                                                   |
| SYM0-T11 | Theorem-relative support integrity | Each support edge has a named theorem, fixed target, supported facts, and denial consequence; the result is relative to the frozen dependency relation.                |
| SYM0-T12 | No second same-scope selector      | After local singleton occupation, fixed-domain gate and apex closure exclude a governance-non-equivalent selector or regime.                                           |
| SYM0-T13 | Total fail-closed disposition      | Every submitted claim reaches one explicit completion or failure disposition; no default normal-form path exists.                                                      |
| SYM0-T14 | Benchmark reconstruction coverage  | The registered constructors reproduce every atomic record in the declared calibration universe; broader candidate generation remains a SYM-11 burden.                  |
| SYM0-T15 | Sixteen-family adequacy            | Every benchmark family is target-fixed, atomized, source-indexed, independently occupied, adversarially tested, and represented without case-specific role vocabulary. |
| SYM0-T16 | Unique symmetry-status normal form | For each complete benchmark target, one lawful assignment orbit survives; every rival is absorbed, excluded, redirected, or theorem-breaking.                          |

## 10.2 Capstone at SYM-0 scope

**Theorem 10.1** (Unique symmetry-status normal form). Fix a target  $T$  in the declared benchmark universe and a role-realization-complete certificate  $C$ . If the conventional-facing physical packet is target-faithful, the candidate-residue quotient is all-and-only presentation collapse, the role architecture is target-derived without occupant identity, all eligibility and compatibility constraints are imposed, and the lawful automorphism group preserves the complete invariant tuple, then

$$|\mathrm{Assign}(C, T) / \Gamma_{\mathrm{lawful}}(C, T)| = 1.$$

The canonical atomic status records are projections of this unique family occupation.

*Proof.* The finite local theorem is Theorem 5.1. Gate uniqueness and no-second-authorizer results exclude an independent same-domain selector after the local orbit has been fixed [23].

Target-update separation removes changed-domain alternatives. The rival protocol exhausts same-orbit, skin, ineligibility, incompatibility, repair, selector import, incomplete certificate, and target-update routes. A remaining same-target second orbit would contradict the verified family occupation theorem and constitutes the explicit falsifier.  $\square$

### 10.3 Exact theorem boundary

The theorem is complete for the declared SYM-0 calibration universe. It does not prove that the sixteen families exhaust every symmetry-relevant structure of the full Standard Model. That candidate-generation and same-scope rival burden is the capstone task of SYM-11. It also does not derive numerical couplings, masses, mixing angles, vacuum scales, or arbitrary ultraviolet uniqueness.

---

## Audit Architecture and Reproducibility

### 11.1 Proof standing and audit standing

The executable project does not create kernel necessity or re-prove imported QFT. It checks the local specialization and its integrity: target and source identity, independent role-position and residue formation, presentation quotient, all-pair eligibility, complete matching search, joint compatibility, automorphism action, orbit decomposition, fail-closed completion, support book-keeping, citation traceability, redescription regressions, manuscript identity, and release integrity.

### 11.2 Controlled proof objects

The release contains twenty-nine immutable target packets, forty-six atomic claims, five hundred forty-seven admitted proposition-level facts, sixteen role architectures, sixty-eight independent physical residues, three hundred eighty-eight eligibility decisions, family-specific compatibility evidence, lawful automorphism registries, assignment-orbit objects, ninety-nine theorem-relative support edges, and one source-to-claim graph. Every public benchmark table is generated from the canonical data.

### 11.3 Independence audit

The dedicated input-independence audit checks that role slots contain no source record, candidate name, semantic subject, or expected occupant; residues contain no final role field or role alias; residues are generated before eligibility; every same-family pair is tested; eligibility uses no semantic-subject equality or role-alias filter; genuine competition exists; and joint compatibility rejects complete preassignments. The frozen metrics are: twenty competing role positions, nineteen multiply eligible residues, twenty-nine complete preassignments, seven substantive rejections, twenty-two compatible assignments, and sixteen singleton orbits.

### 11.4 Adversarial audit

Fourteen mutations exercise the theorem boundary: unsupported fact insertion, source-content mutation, load-bearing fact deletion, missing disposition, extra selector fact, target deletion, support deletion, faithful second-orbit injection, residue omission, stale-atlas substitution, role-position deletion, ineligible pairing, occupant-identity injection, and role-alias injection. Each mutation either fails closed, changes target, preserves the lawful orbit, or explicitly produces the theorem's second-orbit falsifier.

## 11.5 Citation discipline

Every load-bearing citation is typed as provenance, exact support, or context. The evidence graph resolves each admitted fact to a proposition-level source claim with exact location, target assumptions, citation role, persistent identifier, and version authority. Bibliographic metadata and persistent identifiers are verified; the imported proposition remains an explicit premise rather than a machine-derived theorem.

## 11.6 Build reproducibility

The build regenerates the occupation objects, audits certificates and assignments, synchronizes the canonical atlas, runs determinism, redescription, independence, uniqueness, and adversarial checks, generates the manuscript, compiles bibliography and PDF, performs artifact and presentation audits, and verifies the SHA-256 manifest. The formal directory stages the same ontology for subsequent Lean compilation; no machine-checked Lean result is claimed here.

## Conclusion

The Standard Model symmetry problem is not a request to attach optional metaphysical labels to independently determinate QFT objects. Determinate objecthood, same-target comparison, admissible transformation, standing-bearing consequence, and irreversible fixation already force the kernel. Conventional field theory then supplies the faithful sector-specific realization inside that prior governance.

The calculus fixes each target, atomizes each claim, types its physical disposition and causal status, derives role positions from target obligations without naming occupants, forms physical residues before role lookup, evaluates the complete incidence relation, tests every complete matching by substantive joint compatibility, and quotients compatible assignments by the full invariant-preserving automorphism group. The frozen benchmark universe contains genuine competition: ninety eligible edges yield twenty-nine complete preassignments; seven fail joint compatibility; twenty-two remain compatible; and exactly one lawful orbit survives in each of the sixteen families.

The local theorem supplies individual normal-form uniqueness. Fixed-domain gate uniqueness and apex admissible-interior uniqueness then exclude a residual second selector or governance-non-equivalent classification regime. Database labels, expected-occupant names, preselected Boolean vectors, identity hashes, and software repetition play no theorem-bearing role. The result is the surviving orbit of independently formed finite structures under complete eligibility, compatibility, and quotient constraints.

### Publication result

Every role-realization-complete atomic Standard Model symmetry claim in the declared SYM-0 benchmark universe has exactly one admissible role-and-status assignment modulo the full group of lawful identity-preserving redescrptions. No second same-scope selector or untyped role survives.

## Canonical schema and controlled values

The machine-readable schema separates atomic claims, admitted facts, rules, and generated outputs. Load-bearing fields are enumerated or typed composite objects; unrestricted prose is confined to explanations, source locators, and derivation notes.

Table A.1: Canonical atomic atlas summary.

| Record                                 | Target                    | Kind       | Disposition             | Generated roles                             |
|----------------------------------------|---------------------------|------------|-------------------------|---------------------------------------------|
| SYM-00-GAUGE-ARCH-v5                   | D-SM-CORE                 | Structure  | StructuralDatum         | CarrierConstraint                           |
| SYM-00-GAUGE-LOCAL-PROPER-v5           | D-SM-CORE                 | Action     | TrivialAction           | Skin                                        |
| SYM-00-GAUGE-BOUNDARY-RULE-v5          | D-SM-BOUNDARY-RULE        | Diagnostic | ConditionalActionRule   | DerivedWitness                              |
| SYM-00-U1-ASYMPTOTIC-CHARGE-v5         | D-U1-ASYMPTOTIC           | Action     | NontrivialAction        | PhysicalAction                              |
| SYM-00-GAUGE-LARGE-QUOTIENT-v5         | D-YM-LARGE-QUOTIENT       | Action     | TrivialAction           | Skin                                        |
| SYM-00-GAUGE-LARGE-THETA-ACTION-v5     | D-YM-THETA-SECTOR         | Action     | NontrivialAction        | PhysicalAction                              |
| SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5        | D-SM-TOPOLOGY-Z6          | Structure  | StructuralDatum         | Anchor                                      |
| SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 | D-SM-TOPOLOGY-UPDATE      | Structure  | TargetUpdateDisposition | TargetUpdate                                |
| SYM-00-SU2-GLOBAL-ANOMALY-v5           | D-SU2-GLOBAL-ANOMALY-TEST | Diagnostic | DiagnosticNotAction     | DerivedWitness; Obstruction                 |
| SYM-00-BRST-COHOMOLOGY-v5              | D-SM-CORE                 | Action     | CohomologicalProjection | DerivedWitness                              |
| SYM-00-LORENTZ-PASSIVE-v5              | D-SM-CORE                 | Action     | TrivialAction           | Skin                                        |
| SYM-00-POINCARÉ-ACTIVE-v5              | D-SM-CORE                 | Action     | NontrivialAction        | PhysicalAction                              |
| SYM-00-POINCARÉ-ANCHOR-v5              | D-SM-CORE                 | Structure  | StructuralDatum         | Anchor                                      |
| SYM-00-POINCARÉ-NOETHER-WITNESS-v5     | D-SM-CORE                 | Diagnostic | DiagnosticNotAction     | DerivedWitness                              |
| SYM-00-CPT-ACTION-v5                   | D-LOCAL-QFT-CPT           | Action     | NontrivialAction        | PhysicalAction                              |
| SYM-00-CPT-THEOREM-v5                  | D-LOCAL-QFT-CPT           | Diagnostic | DiagnosticNotAction     | DerivedWitness                              |
| SYM-00-U1A-ABJ-DIAGNOSTIC-v5           | D-QCD-MASSLESS-NF3        | Diagnostic | DiagnosticNotAction     | DerivedWitness; Obstruction                 |
| SYM-00-U1A-CLASSICAL-ACTION-v5         | D-QCD-MASSLESS-NF3        | Action     | NontrivialAction        | PhysicalAction; EffectiveRoute; Obstruction |
| SYM-00-U1A-NONINVERTIBLE-ACTION-v5     | D-QED-MASSLESS-DEFECT     | Action     | NontrivialAction        | PhysicalAction; EffectiveRoute              |
| SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5    | D-SM-EW-NONPERT           | Diagnostic | DiagnosticNotAction     | DerivedWitness; Obstruction                 |
| SYM-00-BPLUSL-PERT-ACTION-v5           | D-SM-RENORM-PERT          | Action     | NontrivialAction        | PhysicalAction; EffectiveRoute; Obstruction |
| SYM-00-U35-ZERO-YUKAWA-ACTION-v5       | D-SM-ZERO-YUKAWA          | Action     | NontrivialAction        | PhysicalAction; EffectiveRoute              |
| SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5      | D-SM-QUARK-MIXING         | Diagnostic | DiagnosticNotAction     | DerivedWitness                              |
| SYM-00-MIXING-REPHASING-SKIN-v5        | D-SM-QUARK-MIXING         | Action     | TrivialAction           | Skin                                        |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5  | D-SM-NEUTRINO-UPDATE      | Structure  | TargetUpdateDisposition | TargetUpdate                                |

| Record                                    | Target                  | Kind       | Disposition              | Generated roles                             |
|-------------------------------------------|-------------------------|------------|--------------------------|---------------------------------------------|
| SYM-00-CUSTODIAL-DIAGNOSTIC-v5            | D-EW-CUSTODIAL-LIMIT    | Diagnostic | DiagnosticNotAction      | DerivedWitness; EffectiveRoute              |
| SYM-00-CUSTODIAL-LIMIT-ACTION-v5          | D-EW-CUSTODIAL-LIMIT    | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute              |
| SYM-00-EW-LOCAL-GAUGE-SKIN-v5             | D-EW-HIGGS-PHASE        | Action     | TrivialAction            | Skin                                        |
| SYM-00-EW-VACUUM-FIXATION-v5              | D-EW-HIGGS-PHASE        | Structure  | StructuralDatum          | CarrierConstraint; BridgeFixation           |
| SYM-00-GOLDSTONE-REPRESENTATIVE-v5        | D-EW-HIGGS-PHASE        | Action     | TrivialAction            | Skin                                        |
| SYM-00-U1EM-RESIDUAL-ACTION-v5            | D-EW-HIGGS-PHASE        | Action     | NontrivialAction         | PhysicalAction                              |
| SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5       | D-SM-GLOBAL-FORM-UPDATE | Structure  | TargetUpdateDisposition  | TargetUpdate                                |
| SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5           | D-SM-DEFECT-Z6          | Structure  | StructuralDatum          | Anchor                                      |
| SYM-00-LINE-SPECTRUM-STRUCTURE-v5         | D-SM-DEFECT-Z6          | Structure  | StructuralDatum          | DerivedWitness                              |
| SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 | D-SM-HIGHER-FLAVOR-Z6   | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute; Obstruction |
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5         | D-SM-HIGHER-FLAVOR-Z6   | Structure  | StructuralDatum          | DerivedWitness; EffectiveRoute; Obstruction |
| SYM-00-ONEFORM-ACTION-v5                  | D-SM-DEFECT-Z6          | Action     | NontrivialAction         | PhysicalAction                              |
| SYM-00-EFT-THRESHOLD-MATCHING-v5          | D-SM-EFT-MATCHING-RULE  | Transport  | TheorySpaceTransport     | EffectiveRoute                              |
| SYM-00-FIXED-POINT-SCALE-ACTION-v5        | D-CFT-FIXED-POINT-RULE  | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute              |
| SYM-00-RG-PHYSICAL-FLOW-v5                | D-SM-RG-FLOW            | Transport  | TheorySpaceTransport     | EffectiveRoute                              |
| SYM-00-RG-SCALE-COORDINATE-v5             | D-SM-RG-SCHEME          | Transport  | TheorySpaceRedescription | Skin                                        |
| SYM-00-RG-SCHEME-REDESCRIPTION-v5         | D-SM-RG-SCHEME          | Transport  | TheorySpaceRedescription | Skin                                        |
| SYM-00-SCALE-CLASSICAL-ACTION-v5          | D-SM-MASSLESS-SCALE     | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute; Obstruction |
| SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5        | D-SM-RENORMALIZED-QFT   | Diagnostic | DiagnosticNotAction      | DerivedWitness; Obstruction                 |
| SYM-00-HQET-SPIN-FLAVOR-ACTION-v5         | D-HQET                  | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute              |
| SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5        | D-QCD-ISOSPIN           | Action     | NontrivialAction         | PhysicalAction; EffectiveRoute              |

## A.1 Canonical family occupation maps

### Compact-support local gauge transformation (FAM-01-GAUGE-PROPER)

| Target-derived obligation                      | Function / locus                       | Independently formed occupying residue                                                                   |
|------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-01-GAUGE-PROPER-01-CARRIERCONSTRAINT | CarrierConstraint; Architecture/Target | RES7-sym-00-gauge-arch-v5-architecture-dependency; architecture_dependency; architecture_denial          |
| SLOT7-FAM-01-GAUGE-PROPER-02-SKIN              | Skin; Representative/Bulk              | RES7-sym-00-gauge-local-proper-v5-presentation-relation; presentation_relation; presentation_equivalence |

### Gauge transformation nontrivial at a boundary (FAM-02-GAUGE-BOUNDARY)

| Target-derived obligation                     | Function / locus                     | Independently formed occupying residue                                                            |
|-----------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------|
| SLOT7-FAM-02-GAUGE-BOUNDARY-01-DERIVEDWITNESS | DerivedWitness; Consequence/Boundary | RES7-sym-00-gauge-boundary-rule-v5-diagnostic-content; diagnostic_content; diagnostic_consequence |
| SLOT7-FAM-02-GAUGE-BOUNDARY-02-PHYSICALACTION | PhysicalAction; Boundary/Boundary    | RES7-sym-00-u1-asymptotic-charge-v5-action-object; action_object; quotient_action                 |

### Large gauge transformation and theta sector (FAM-03-GAUGE-LARGE)

| Target-derived obligation                  | Function / locus                               | Independently formed occupying residue                                                                     |
|--------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-03-GAUGE-LARGE-01-SKIN           | Skin; Representative/Topology                  | RES7-sym-00-gauge-large-quotient-v5-presentation-relation; presentation_relation; presentation_equivalence |
| SLOT7-FAM-03-GAUGE-LARGE-02-PHYSICALACTION | PhysicalAction; PhysicalAction/StateSpace      | RES7-sym-00-gauge-large-theta-action-v5-action-object; action_object; quotient_action                      |
| SLOT7-FAM-03-GAUGE-LARGE-03-ANCHOR         | Anchor; TargetTyping/Target                    | RES7-sym-00-gauge-topology-anchor-v5-target-definition; target_definition; target_definition               |
| SLOT7-FAM-03-GAUGE-LARGE-04-TARGETUPDATE   | TargetUpdate; TargetComparison/Target          | RES7-sym-00-gauge-topology-target-update-v5-target-comparison; target_comparison; target_difference        |
| SLOT7-FAM-03-GAUGE-LARGE-05-DERIVEDWITNESS | DerivedWitness; Consequence/Topology           | RES7-sym-00-su2-global-anomaly-v5-diagnostic-content; diagnostic_content; diagnostic_consequence           |
| SLOT7-FAM-03-GAUGE-LARGE-06-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/GlobalGauge | RES7-sym-00-su2-global-anomaly-v5-consistency-condition; consistency_condition; failure_condition          |

**BRST symmetry (FAM-04-BRST)**

| Target-derived obligation           | Function / locus                       | Independently formed occupying residue                                                                   |
|-------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-04-BRST-01-DERIVEDWITNESS | DerivedWitness; Consequence/FieldSpace | RES7-sym-00-brst-cohomology-v5-cohomological-relation; cohomological_relation; cohomological_consequence |

**Poincaré transformation (FAM-05-POINCARE)**

| Target-derived obligation               | Function / locus                           | Independently formed occupying residue                                                                 |
|-----------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-05-POINCARE-01-SKIN           | Skin; Representative/FieldSpace            | RES7-sym-00-lorentz-passive-v5-presentation-relation; presentation_relation; presentation_equivalence  |
| SLOT7-FAM-05-POINCARE-02-PHYSICALACTION | PhysicalAction; PhysicalAction/StateSpace  | RES7-sym-00-poincare-active-v5-action-object; action_object; quotient_action                           |
| SLOT7-FAM-05-POINCARE-03-ANCHOR         | Anchor; TargetTyping/Target                | RES7-sym-00-poincare-anchor-v5-target-definition; target_definition; target_definition                 |
| SLOT7-FAM-05-POINCARE-04-DERIVEDWITNESS | DerivedWitness; Consequence/LocalOperators | RES7-sym-00-poincare-noether-witness-v5-diagnostic-content; diagnostic_content; diagnostic_consequence |

**CPT (FAM-06-CPT)**

| Target-derived obligation          | Function / locus                          | Independently formed occupying residue                                                    |
|------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------|
| SLOT7-FAM-06-CPT-01-PHYSICALACTION | PhysicalAction; PhysicalAction/StateSpace | RES7-sym-00-cpt-action-v5-action-object; action_object; quotient_action                   |
| SLOT7-FAM-06-CPT-02-DERIVEDWITNESS | DerivedWitness; Consequence/Target        | RES7-sym-00-cpt-theorem-v5-diagnostic-content; diagnostic_content; diagnostic_consequence |

### Axial U(1)<sub>A</sub> (FAM-07-U1A)

| Target-derived obligation              | Function / locus                                           | Independently formed occupying residue                                                                             |
|----------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-07-U1A-01-D<br>ERIVEDWITNESS | DerivedWitness; Con-<br>sequence/LocalOpera-<br>tors       | RES7-sym-00-u1a-abj-diagno-<br>stic-v5-diagnostic-conten-<br>t; diagnostic_content; diagnos-<br>tic_consequence    |
| SLOT7-FAM-07-U1A-02-0<br>BSTRUCTION    | Obstruction; Con-<br>sistencyOrMatch-<br>ing/ABJ           | RES7-sym-00-u1a-abj-diagno-<br>stic-v5-consistency-condit-<br>ion; consistency_condition; fail-<br>ure_condition   |
| SLOT7-FAM-07-U1A-03-P<br>HYSICALACTION | PhysicalAction; Physi-<br>calAction/FieldSpace             | RES7-sym-00-u1a-classical-<br>action-v5-action-object; ac-<br>tion_object; quotient_action                         |
| SLOT7-FAM-07-U1A-04-E<br>FFECTIVEROUTE | EffectiveRoute;<br>Regime/Parameter-<br>Limit              | RES7-sym-00-u1a-classic-<br>al-action-v5-regime-co-<br>ndition; regime_condition;<br>regime_condition              |
| SLOT7-FAM-07-U1A-05-0<br>BSTRUCTION    | Obstruction; Con-<br>sistencyOrMatch-<br>ing/ABJ           | RES7-sym-00-u1a-classical-a-<br>ction-v5-consistency-condit-<br>ion; consistency_condition; fail-<br>ure_condition |
| SLOT7-FAM-07-U1A-06-P<br>HYSICALACTION | PhysicalAction; Physi-<br>calAction/Extended-<br>Operators | RES7-sym-00-u1a-noninvertib-<br>le-action-v5-action-object;<br>action_object; quotient_action                      |
| SLOT7-FAM-07-U1A-07-E<br>FFECTIVEROUTE | EffectiveRoute;<br>Regime/Parameter-<br>Limit              | RES7-sym-00-u1a-noninver-<br>tible-action-v5-regime-c-<br>ondition; regime_condition;<br>regime_condition          |

### Electroweak B+L (FAM-08-BPLUSL)

| Target-derived obligation                 | Function / locus                                               | Independently formed occupying residue                                                                                 |
|-------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-08-BPLUSL-0<br>1-DERIVEDWITNESS | DerivedWitness; Con-<br>sequence/Topology                      | RES7-sym-00-bplusl-nonpert-d-<br>iagnostic-v5-diagnostic-cont-<br>ent; diagnostic_content; diagnos-<br>tic_consequence |
| SLOT7-FAM-08-BPLUSL-0<br>2-OBSTRUCTION    | Obstruction; Consis-<br>tencyOrMatching/-<br>ElectroweakBPlusL | RES7-sym-00-bplusl-nonpert-<br>diagnostic-v5-consistency-c-<br>ondition; consistency_condition;<br>failure_condition   |
| SLOT7-FAM-08-BPLUSL-0<br>3-PHYSICALACTION | PhysicalAction; Physi-<br>calAction/StateSpace                 | RES7-sym-00-bplusl-pert-a-<br>ction-v5-action-object; ac-<br>tion_object; quotient_action                              |

| Target-derived obligation             | Function / locus                                      | Independently formed occupying residue                                                                         |
|---------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-08-BPLUSL-04-EFFECTIVEROUTE | EffectiveRoute; Regime/OperatorDimension              | RES7-sym-00-bplusl-per<br>t-action-v5-regime-con<br>dition; regime_condition;<br>regime_condition              |
| SLOT7-FAM-08-BPLUSL-05-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/-ElectroweakBPlusL | RES7-sym-00-bplusl-pert-ac<br>tion-v5-consistency-condit<br>ion; consistency_condition; fail-<br>ure_condition |

### U(3)<sup>5</sup> flavor limit (FAM-09-U35)

| Target-derived obligation          | Function / locus                          | Independently formed occupying residue                                                                |
|------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-09-U35-01-PHYSICALACTION | PhysicalAction; PhysicalAction/FieldSpace | RES7-sym-00-u35-zero-yukaw<br>a-action-v5-action-object;<br>action_object; quotient_action            |
| SLOT7-FAM-09-U35-02-EFFECTIVEROUTE | EffectiveRoute; Regime/Parameter-Limit    | RES7-sym-00-u35-zero-yu<br>kawa-action-v5-regime-c<br>ondition; regime_condition;<br>regime_condition |

### CKM/PMNS rephasing (FAM-10-MIXING)

| Target-derived obligation             | Function / locus                           | Independently formed occupying residue                                                                                   |
|---------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-10-MIXING-01-DERIVEDWITNESS | DerivedWitness; Consequence/ParameterSpace | RES7-sym-00-ckm-jarlskog-di<br>agnostic-v5-diagnostic-cont<br>ent; diagnostic_content; diagnos-<br>tic_consequence       |
| SLOT7-FAM-10-MIXING-02-SKIN           | Skin; Representative/-FieldSpace           | RES7-sym-00-mixing-rephasing-<br>skin-v5-presentation-relatio<br>n; presentation_relation; presenta-<br>tion_equivalence |
| SLOT7-FAM-10-MIXING-03-TARGETUPDATE   | TargetUpdate; Target-Comparison/Target     | RES7-sym-00-pmns-neutrino-t<br>arget-update-v5-target-comp<br>arison; target_comparison; tar-<br>get_difference          |

### Custodial symmetry (FAM-11-CUSTODIAL)

| Target-derived obligation                | Function / locus                           | Independently formed occupying residue                                                             |
|------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------|
| SLOT7-FAM-11-CUSTODIAL-01-DERIVEDWITNESS | DerivedWitness; Consequence/LocalOperators | RES7-sym-00-custodial-diagnostic-v5-diagnostic-content; diagnostic_content; diagnostic_consequence |
| SLOT7-FAM-11-CUSTODIAL-02-EFFECTIVEROUTE | EffectiveRoute; Regime/Parameter-Limit     | RES7-sym-00-custodial-diagnostic-v5-regime-condition; regime_condition; regime_condition           |
| SLOT7-FAM-11-CUSTODIAL-03-PHYSICALACTION | PhysicalAction; PhysicalAction/Vacuum      | RES7-sym-00-custodial-limit-action-v5-action-object; action_object; quotient_action                |
| SLOT7-FAM-11-CUSTODIAL-04-EFFECTIVEROUTE | EffectiveRoute; Regime/Parameter-Limit     | RES7-sym-00-custodial-limit-action-v5-regime-condition; regime_condition; regime_condition         |

### Electroweak gauge breaking (FAM-12-EW-FIXATION)

| Target-derived obligation                     | Function / locus                          | Independently formed occupying residue                                                                         |
|-----------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-12-EW-FIXATION-01-SKIN              | Skin; Representative/Bulk                 | RES7-sym-00-ew-local-gauge-skin-v5-presentation-relation; presentation_relation; presentation_equivalence      |
| SLOT7-FAM-12-EW-FIXATION-02-CARRIERCONSTRAINT | CarrierConstraint; Architecture/Vacuum    | RES7-sym-00-ew-vacuum-fixation-v5-architecture-dependency; architecture_dependency; architecture_denial        |
| SLOT7-FAM-12-EW-FIXATION-03-BRIDGEFIXATION    | BridgeFixation; Transition/Vacuum         | RES7-sym-00-ew-vacuum-fixation-v5-transition-relation; transition_relation; transition_mediation               |
| SLOT7-FAM-12-EW-FIXATION-04-SKIN              | Skin; Representative/-FieldSpace          | RES7-sym-00-goldstone-representative-v5-presentation-relation; presentation_relation; presentation_equivalence |
| SLOT7-FAM-12-EW-FIXATION-05-PHYSICALACTION    | PhysicalAction; PhysicalAction/StateSpace | RES7-sym-00-u1em-residual-action-v5-action-object; action_object; quotient_action                              |

**G\_SM/Gamma (FAM-13-GLOBAL-FORM)**

| Target-derived obligation                  | Function / locus                               | Independently formed occupying residue                                                                        |
|--------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-13-GLOBAL-FORM-01-TARGETUPDATE   | TargetUpdate; Target-Comparison/Target         | RES7-sym-00-global-form-target-update-v5-target-comparison; target_comparison; target_difference              |
| SLOT7-FAM-13-GLOBAL-FORM-02-ANCHOR         | Anchor; TargetTyping/Target                    | RES7-sym-00-global-form-z6-anchor-v5-target-definition; target_definition; target_definition                  |
| SLOT7-FAM-13-GLOBAL-FORM-03-DERIVEDWITNESS | DerivedWitness; Consequence/Extended-Operators | RES7-sym-00-line-spectrum-structure-v5-structural-consequence; structural_consequence; dependency_consequence |

**One-form/higher-group flavor structure (FAM-14-GENERALIZED)**

| Target-derived obligation                  | Function / locus                                  | Independently formed occupying residue                                                                         |
|--------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-14-GENERALIZED-01-PHYSICALACTION | PhysicalAction; PhysicalAction/Extended-Operators | RES7-sym-00-higher-flavor-background-action-v5-action-object; action_object; quotient_action                   |
| SLOT7-FAM-14-GENERALIZED-02-EFFECTIVEROUTE | EffectiveRoute; Regime/Parameter-Limit            | RES7-sym-00-higher-flavor-background-action-v5-regime-condition; regime_condition; regime_condition            |
| SLOT7-FAM-14-GENERALIZED-03-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/tHooft         | RES7-sym-00-higher-flavor-background-action-v5-consistency-condition; consistency_condition; failure_condition |
| SLOT7-FAM-14-GENERALIZED-04-DERIVEDWITNESS | DerivedWitness; Consequence/Target                | RES7-sym-00-higher-flavor-extension-v5-structural-consequence; structural_consequence; dependency_consequence  |
| SLOT7-FAM-14-GENERALIZED-05-EFFECTIVEROUTE | EffectiveRoute; Regime/Parameter-Limit            | RES7-sym-00-higher-flavor-extension-v5-regime-condition; regime_condition; regime_condition                    |
| SLOT7-FAM-14-GENERALIZED-06-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/tHooft         | RES7-sym-00-higher-flavor-extension-v5-consistency-condition; consistency_condition; failure_condition         |
| SLOT7-FAM-14-GENERALIZED-07-PHYSICALACTION | PhysicalAction; PhysicalAction/Extended-Operators | RES7-sym-00-oneform-action-v5-action-object; action_object; quotient_action                                    |

## Classical scale symmetry and trace anomaly (FAM-15-SCALE-RG)

| Target-derived obligation               | Function / locus                              | Independently formed occupying residue                                                                        |
|-----------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| SLOT7-FAM-15-SCALE-RG-01-EFFECTIVEROUTE | EffectiveRoute;<br>Regime/EFTOrder            | RES7-sym-00-eft-threshold-matching-v5-transport-relation; transport_relation; transport                       |
| SLOT7-FAM-15-SCALE-RG-02-PHYSICALACTION | PhysicalAction; PhysicalAction/LocalOperators | RES7-sym-00-fixed-point-scale-action-v5-action-object; action_object; quotient_action                         |
| SLOT7-FAM-15-SCALE-RG-03-EFFECTIVEROUTE | EffectiveRoute;<br>Regime/FixedPoint          | RES7-sym-00-fixed-point-scale-action-v5-regime-condition; regime_condition; regime_condition                  |
| SLOT7-FAM-15-SCALE-RG-04-EFFECTIVEROUTE | EffectiveRoute;<br>Regime/Matching-Interval   | RES7-sym-00-rg-physical-flow-v5-transport-relation; transport_relation; transport                             |
| SLOT7-FAM-15-SCALE-RG-05-SKIN           | Skin; Representative/TheorySpace              | RES7-sym-00-rg-scale-coordinate-v5-presentation-relation; presentation_relation; presentation_equivalence     |
| SLOT7-FAM-15-SCALE-RG-06-SKIN           | Skin; Representative/TheorySpace              | RES7-sym-00-rg-scheme-redescription-v5-presentation-relation; presentation_relation; presentation_equivalence |
| SLOT7-FAM-15-SCALE-RG-07-PHYSICALACTION | PhysicalAction; PhysicalAction/FieldSpace     | RES7-sym-00-scale-classical-action-v5-action-object; action_object; quotient_action                           |
| SLOT7-FAM-15-SCALE-RG-08-EFFECTIVEROUTE | EffectiveRoute;<br>Regime/Parameter-Limit     | RES7-sym-00-scale-classical-action-v5-regime-condition; regime_condition; regime_condition                    |
| SLOT7-FAM-15-SCALE-RG-09-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/-Trace     | RES7-sym-00-scale-classical-action-v5-consistency-condition; consistency_condition; failure_condition         |
| SLOT7-FAM-15-SCALE-RG-10-DERIVEDWITNESS | DerivedWitness; Consequence/LocalOperators    | RES7-sym-00-trace-anomaly-diagnostic-v5-diagnostic-content; diagnostic_content; diagnostic_consequence        |
| SLOT7-FAM-15-SCALE-RG-11-OBSTRUCTION    | Obstruction; ConsistencyOrMatching/-Trace     | RES7-sym-00-trace-anomaly-diagnostic-v5-consistency-condition; consistency_condition; failure_condition       |

### Isospin or heavy-quark symmetry (FAM-16-EFFECTIVE)

| Target-derived obligation                | Function / locus                            | Independently formed occupying residue                                                       |
|------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------|
| SLOT7-FAM-16-EFFECTIVE-01-PHYSICALACTION | PhysicalAction; PhysicalAction/StateSpace   | RES7-sym-00-hqet-spin-flavor-action-v5-action-object; action_object; quotient_action         |
| SLOT7-FAM-16-EFFECTIVE-02-EFFECTIVEROUTE | EffectiveRoute; Regime/Asymptotic-Expansion | RES7-sym-00-hqet-spin-flavor-action-v5-regime-condition; regime_condition; regime_condition  |
| SLOT7-FAM-16-EFFECTIVE-03-PHYSICALACTION | PhysicalAction; PhysicalAction/StateSpace   | RES7-sym-00-isospin-effective-action-v5-action-object; action_object; quotient_action        |
| SLOT7-FAM-16-EFFECTIVE-04-EFFECTIVEROUTE | EffectiveRoute; Regime/IRRegime             | RES7-sym-00-isospin-effective-action-v5-regime-condition; regime_condition; regime_condition |

# Targets, constructors, and family partitions

Table B.1: Immutable target registry.

| Target ID                 | Scope class        | Fixed distinguishing data                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-SM-CORE                 | core               | Sharp-local renormalizable Standard Model target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix.. |
| D-SM-BOUNDARY-RULE        | conditional_rule   | Boundary gauge-action rule target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Bulk observables plus boundary observables and surface charges..                                        |
| D-U1-ASYMPTOTIC           | controlled_variant | Asymptotic U(1) charge target; topology: Topologically trivial U(1) bundle sector.; global form: U(1).; operator category: Bulk gauge-invariant observables, charged asymptotic states, and the electric surface charge..                                                                                                                                                                                            |
| D-YM-LARGE-QUOTIENT       | controlled_variant | Large transformations included in the gauge quotient; topology: Maps $S^3 \rightarrow SU(2)$ of every winding number are included in the gauge-equivalence relation.; global form: $SU(2)$ .; operator category: Gauge-orbit state space and local gauge-invariant operators..                                                                                                                                       |
| D-YM-THETA-SECTOR         | controlled_variant | Theta-represented large-gauge target; topology: Maps $S^3 \rightarrow SU(2)$ have integer winding; the quantum state carries the one-dimensional representation $U_n \theta\rangle = \exp(in\theta) \theta\rangle$ ; global form: $SU(2)$ .; operator category: Wave functionals, theta sectors, and local gauge-invariant operators..                                                                               |
| D-SM-TOPOLOGY-UPDATE      | target_update      | Topology-completed Standard Model target update; topology: The candidate changes the sharp-local core by admitting nontrivial principal bundles, winding sectors, and instanton number.; global form: A faithful global gauge group must be selected in the updated target.; operator category: Partition functions, topological sectors, and extended operators..                                                   |
| D-SM-TOPOLOGY-Z6          | controlled_variant | Topology-completed Z6 Standard Model target; topology: Principal bundles compatible with $(SU(3) \times SU(2) \times U(1))/Z_6$ , including nonzero topological sectors.; global form: $(SU(3) \times SU(2) \times U(1))/Z_6$ .; operator category: Local operators, partition functions, instanton sectors, and genuine extended operators..                                                                        |
| D-SU2-GLOBAL-ANOMALY-TEST | worked_model       | Witten SU(2) global-anomaly test target; topology: Gauge transformations include the nontrivial class of $\pi_4(SU(2)) = \mathbb{Z}_2$ ; global form: $SU(2)$ .; operator category: Fermion determinant and partition function..                                                                                                                                                                                     |
| D-LOCAL-QFT-CPT           | controlled_variant | Local QFT CPT premise target; topology: Topologically trivial spacetime sector.; global form: NotApplicable; operator category: Local fields, states, and the S-matrix..                                                                                                                                                                                                                                             |

| Target ID               | Scope class        | Fixed distinguishing data                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-QCD-MASSLESS-NF3      | controlled_variant | Three-flavor massless QCD target; topology: Instanton sectors admitted.; global form: $SU(3)$ .; operator category: Local operators, hadron states, topological sectors, and chiral defects where explicitly stated..                                                                                                                                                                                                                                                        |
| D-QED-MASSLESS-DEFECT   | controlled_variant | Massless QED non-invertible chiral-defect target; topology: Magnetic line sectors admitted.; global form: $U(1)$ , with genuine t Hooft lines admitted and no dynamical magnetic monopoles.; operator category: Local operators, t Hooft lines, and topological defects..                                                                                                                                                                                                    |
| D-SM-RENORM-PERT        | core               | Perturbative renormalizable Standard Model target; topology: Topologically trivial sector; electroweak instanton and sphaleron sectors excluded from this child target.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix.. |
| D-SM-EW-NONPERT         | core               | Nonperturbative electroweak Standard Model target; topology: Electroweak instanton number and sphaleron transitions are admitted.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..                                       |
| D-SM-ZERO-YUKAWA        | core               | Zero-Yukawa flavor-limit target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..                                                                          |
| D-SM-QUARK-MIXING       | core               | Three-generation quark-mixing target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..                                                                     |
| D-SM-NEUTRINO-UPDATE    | target_update      | Neutrino-sector target update family; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..                                                                     |
| D-EW-CUSTODIAL-LIMIT    | core               | Electroweak custodial-symmetry limit; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..                                                                     |
| D-EW-HIGGS-PHASE        | core               | Gauge-invariant one-doublet electroweak Higgs-phase target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local/composite operators, physical charged sectors, and asymptotic particle observables..                             |
| D-SM-GLOBAL-FORM-UPDATE | target_update      | Faithful global-form target update; topology: Topologically trivial principal-bundle sector only.; global form: The candidate replaces the sharp-local target by one member of the family $(SU(3) \times SU(2) \times U(1))/\Gamma$ with $ \Gamma  \leq Z_6$ .; operator category: The update admits genuine Wilson, t Hooft, and dyonic line operators..                                                                                                                    |

| Target ID              | Scope class        | Fixed distinguishing data                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-SM-DEFECT-Z6         | controlled_variant | Z6 global-form defect-completed Standard Model target; topology: Principal bundles compatible with the Z6 quotient; no dynamical magnetic monopoles.; global form: $(SU(3) \times SU(2) \times U(1))/Z_6$ .; operator category: Local operators plus genuine Wilson, t Hooft, and dyonic line operators; screening by Standard Model matter is included..                                                             |
| D-SM-HIGHER-FLAVOR-Z6  | controlled_variant | Higher-flavor background-coupled Z6 Standard Model target; topology: Z6-compatible bundle sectors; no dynamical hypercharge monopoles.; global form: $(SU(3) \times SU(2) \times U(1))/Z_6$ with magnetic hypercharge one-form background.; operator category: Local operators, genuine lines, zero-form flavor backgrounds, and a magnetic one-form background two-form field..                                      |
| D-SM-MASSLESS-SCALE    | core               | Classically scale-invariant Standard Model limit; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Gauge-invariant local operators, physical asymptotic particle states, and the S-matrix..  |
| D-SM-RENORMALIZED-QFT  | core               | Renormalized Standard Model operator target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Renormalized local composite operators, the improved stress tensor, and physical correlators.. |
| D-SM-RG-SCHEME         | core               | Analytically related renormalization schemes target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Couplings, renormalized operators, amplitudes, and the analytic scheme map..           |
| D-SM-RG-FLOW           | core               | Standard Model RG trajectory target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: Theory-space points, running couplings, renormalized operators, and scale-dependent amplitudes..       |
| D-SM-EFT-MATCHING-RULE | conditional_rule   | EFT threshold-matching rule target; topology: Topologically trivial principal-bundle sector only.; global form: Sharp-local target: the local gauge Lie algebra and observed representations are fixed; extended line operators and faithful quotient data are excluded from this target.; operator category: UV and IR operators, Wilson coefficients, and matched amplitudes..                                      |
| D-CFT-FIXED-POINT-RULE | conditional_rule   | Fixed-point scale-action rule target; topology: Topologically trivial spacetime sector.; global form: Fixed by the antecedent.; operator category: Local operators, states, and fixed-point correlators..                                                                                                                                                                                                             |
| D-QCD-ISOSPIN          | controlled_variant | Two-flavor isospin target; topology: QCD topological sectors admitted.; global form: $SU(3)$ .; operator category: Gauge-invariant local operators and hadron states..                                                                                                                                                                                                                                                |
| D-HQET                 | controlled_variant | Leading-order HQET target; topology: QCD topological sectors admitted.; global form: $SU(3)$ .; operator category: HQET operators and heavy-hadron states..                                                                                                                                                                                                                                                           |

Table B.2: Implemented constructor registry.

| ID                       | Kind      | Disposition            | Function                                                 |
|--------------------------|-----------|------------------------|----------------------------------------------------------|
| CTR-TARGET-ARCH          | Structure | <b>StructuralDatum</b> | Target-bearing architecture constructor                  |
| CTR-PROPER-REDESCRIPTION | Action    | <b>TrivialAction</b>   | Proper gauge/basis/coordinate representative constructor |

| ID                            | Kind       | Disposition              | Function                                          |
|-------------------------------|------------|--------------------------|---------------------------------------------------|
| CTR-BOUNDARY-RULE             | Diagnostic | ConditionalActionRule    | Conditional boundary-action rule constructor      |
| CTR-PHYSICAL-ACTION           | Action     | NontrivialAction         | Quotient-nontrivial action constructor            |
| CTR-LARGE-QUOTIENT            | Action     | TrivialAction            | Large transformation included in quotient         |
| CTR-TOPOLOGY-UPDATE           | Structure  | TargetUpdateDisposition  | Topology/global-form target-update constructor    |
| CTR-DIAGNOSTIC-OBSTRUCTION    | Diagnostic | DiagnosticNotAction      | Anomaly or failure diagnostic constructor         |
| CTR-COHOMOLOGICAL             | Action     | CohomologicalProjection  | BRST/cohomological constructor                    |
| CTR-DOMAIN-ANCHOR             | Structure  | StructuralDatum          | Domain-anchor constructor                         |
| CTR-NOETHER-DIAGNOSTIC        | Diagnostic | DiagnosticNotAction      | Noether or invariant witness constructor          |
| CTR-DERIVED-THEOREM           | Diagnostic | DiagnosticNotAction      | Derived theorem consequence constructor           |
| CTR-LIMIT-ACTION              | Action     | NontrivialAction         | Parameter-limit physical action constructor       |
| CTR-TARGET-UPDATE             | Structure  | TargetUpdateDisposition  | Declared target-update constructor                |
| CTR-BRIDGE-FIXATION           | Structure  | StructuralDatum          | Bridge/fixation constructor                       |
| CTR-STRUCTURAL-WITNESS        | Structure  | StructuralDatum          | Structural datum derived from target architecture |
| CTR-GENERALIZED-ACTION        | Action     | NontrivialAction         | Extended-operator or defect action constructor    |
| CTR-THEORYSPACE-REDESCRIPTION | Transport  | TheorySpaceRedescription | Scheme/coordinate redescription constructor       |
| CTR-THEORYSPACE-TRANSPORT     | Transport  | TheorySpaceTransport     | Physical RG/EFT transport constructor             |
| CTR-FIXEDPOINT-RULE           | Action     | NontrivialAction         | Conditional fixed-point action constructor        |

Table B.3: Parent-family partition registry.

| Certificate                 | Family                                        | Atomic children                                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PART5-FAM-01-GAUGE-PROPER   | Compact-support local gauge transformation    | SYM-00-GAUGE-ARCH-v5, SYM-00-GAUGE-LOCAL-PROPER-v5                                                                                                                                                                                       |
| PART5-FAM-02-GAUGE-BOUNDARY | Gauge transformation nontrivial at a boundary | SYM-00-GAUGE-BOUNDARY-RULE-v5, SYM-00-U1-ASYMPTOTIC-CHARGE-v5                                                                                                                                                                            |
| PART5-FAM-03-GAUGE-LARGE    | Large gauge transformation and theta sector   | SYM-00-GAUGE-LARGE-QUOTIENT-v5, SYM-00-GAUGE-LARGE-THETA-ACTION-v5, SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5, SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5, SYM-00-SU2-GLOBAL-ANOMALY-v5                                                                |
| PART5-FAM-04-BRST           | BRST symmetry                                 | SYM-00-BRST-COHOMOLOGY-v5                                                                                                                                                                                                                |
| PART5-FAM-05-POINCARÉ       | Poincaré transformation                       | SYM-00-LORENTZ-PASSIVE-v5, SYM-00-POINCARÉ-ACTIVE-v5, SYM-00-POINCARÉ-ANCHOR-v5, SYM-00-POINCARÉ-NOETHER-WITNESS-v5                                                                                                                      |
| PART5-FAM-06-CPT            | CPT                                           | SYM-00-CPT-ACTION-v5, SYM-00-CPT-THEOREM-v5                                                                                                                                                                                              |
| PART5-FAM-07-U1A            | Axial U(1) <sub>A</sub>                       | SYM-00-U1A-ABJ-DIAGNOSTIC-v5, SYM-00-U1A-CLASSICAL-ACTION-v5, SYM-00-U1A-NONINVERTIBLE-ACTION-v5                                                                                                                                         |
| PART5-FAM-08-BPLUSL         | Electroweak B+L                               | SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5, SYM-00-BPLUSL-PERT-ACTION-v5                                                                                                                                                                        |
| PART5-FAM-09-U35            | U(3) <sup>5</sup> flavor limit                | SYM-00-U35-ZERO-YUKAWA-ACTION-v5                                                                                                                                                                                                         |
| PART5-FAM-10-MIXING         | CKM/PMNS rephasing                            | SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5, SYM-00-MIXING-REPHASING-SKIN-v5, SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5                                                                                                                                |
| PART5-FAM-11-CUSTODIAL      | Custodial symmetry                            | SYM-00-CUSTODIAL-DIAGNOSTIC-v5, SYM-00-CUSTODIAL-LIMIT-ACTION-v5                                                                                                                                                                         |
| PART5-FAM-12-EW-FIXATION    | Electroweak gauge breaking                    | SYM-00-EW-LOCAL-GAUGE-SKIN-v5, SYM-00-EW-VACUUM-FIXATION-v5, SYM-00-GOLDSTONE-REPRESENTATIVE-v5, SYM-00-U1EM-RESIDUAL-ACTION-v5                                                                                                          |
| PART5-FAM-13-GLOBAL-FORM    | G <sub>SM</sub> /Gamma                        | SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5, SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5, SYM-00-LINE-SPECTRUM-STRUCTURE-v5                                                                                                                                  |
| PART5-FAM-14-GENERALIZED    | One-form/higher-group flavor structure        | SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5, SYM-00-HIGHER-FLAVOR-EXTENSION-v5, SYM-00-ONEFORM-ACTION-v5                                                                                                                                   |
| PART5-FAM-15-SCALE-RG       | Classical scale symmetry and trace anomaly    | SYM-00-EFT-THRESHOLD-MATCHING-v5, SYM-00-FIXED-POINT-SCALE-ACTION-v5, SYM-00-RG-PHYSICAL-FLOW-v5, SYM-00-RG-SCALE-COORDINATE-v5, SYM-00-RG-SCHEME-REDESCRIPTION-v5, SYM-00-SCALE-CLASSICAL-ACTION-v5, SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5 |
| PART5-FAM-16-EFFECTIVE      | Isospin or heavy-quark symmetry               | SYM-00-HQET-SPIN-FLAVOR-ACTION-v5, SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5                                                                                                                                                                    |

## Support and source traceability

Table C.1: Support graph by carrier.

| Carrier           | Edges | Removal-audit interpretation                                                                                                 |
|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------|
| K-ANOMALY-FREE    | 4     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-BOUNDARY-DATA   | 2     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-DEFECT-CATEGORY | 6     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-EFT-MATCHING    | 3     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-EW-TOPOLOGY     | 1     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-GAUGE-ORDER     | 11    | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-GLOBAL-FORM     | 8     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-HIGGS-BRIDGE    | 5     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-LOCALITY        | 3     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-MATTER-REP      | 12    | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-MINKOWSKI       | 8     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |

| Carrier           | Edges | Removal-audit interpretation                                                                                                 |
|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------|
| K-QCD-CHIRAL      | 3     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-QCD-DYNAMICS    | 3     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-QUANTIZATION    | 4     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-REFERENCE       | 1     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-RG-TRANSPORT    | 7     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-TOPOLOGY        | 6     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-UNITARITY       | 3     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-VACUUM-FIXATION | 4     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |
| K-YUKAWA          | 5     | Each retained edge names an imported necessity theorem, held-fixed target fields, and a distinct cost-of-denial consequence. |

Table C.2: Source authority and claim traceability.

| Source                      | Claims | Persistent identifier          | Verification basis                  |
|-----------------------------|--------|--------------------------------|-------------------------------------|
| Adler1969                   | 24     | 10.1103/PhysRev.177.2426       | Bibliographic web audit             |
| AharonySeibergTachikawa2013 | 41     | 10.1007/JHEP08(2013)115        | Official arXiv record               |
| CallanColemanJackiw1970     | 67     | 10.1016/0003-4916(70)90394-5   | Bibliographic web audit             |
| ChoiLamShao2022             | 13     | 10.1103/PhysRevLett.129.161601 | Official arXiv record               |
| CordovaKoren2023            | 29     | 10.1002/andp.202300031         | Official arXiv record               |
| Elitzur1975                 | 50     | 10.1103/PhysRevD.12.3978       | Bibliographic web audit             |
| GasserLeutwyler1984         | 11     | 10.1016/0003-4916(84)90242-2   | Bibliographic web audit             |
| GreavesThomas2014           | 18     | 10.1016/j.shpsb.2013.10.001    | Official arXiv record               |
| HenneauxTeitelboim1992      | 97     | Frozen snapshot                | Bibliographic web audit             |
| Jarlskog1985                | 39     | 10.1103/PhysRevLett.55.1039    | Bibliographic web audit             |
| MaleyAER                    | 2      | Frozen snapshot                | Frozen snapshot authority           |
| MaleyFixedDomain            | 2      | 10.5281/zenodo.18682423        | Author DOI ledger + frozen snapshot |
| MaleyRPI                    | 1      | 10.5281/zenodo.20129729        | Author DOI ledger + frozen snapshot |
| MaleyRoleOccupancy          | 33     | Frozen snapshot                | Frozen snapshot authority           |
| MaleySM2                    | 15     | 10.5281/zenodo.21213644        | Author DOI ledger + frozen snapshot |
| MaleySM4                    | 16     | 10.5281/zenodo.21213845        | Author DOI ledger + frozen snapshot |
| MaleySM5                    | 8      | 10.5281/zenodo.21214028        | Author DOI ledger + frozen snapshot |
| MaleyStandaloneSM           | 2      | 10.5281/zenodo.20305720        | Author DOI ledger + frozen snapshot |
| Neubert1994                 | 12     | 10.1016/0370-1573(94)90091-4   | Official arXiv record               |
| Wigner1939                  | 33     | 10.2307/1968551                | Bibliographic web audit             |
| Witten1982                  | 10     | 10.1016/0370-2693(82)90728-6   | Bibliographic web audit             |
| tHooft1976                  | 24     | 10.1103/PhysRevLett.37.8       | Bibliographic web audit             |

# Standing-Filtered Hostile-Referee Disposition

## D.1 Dependency order

The demand that AASC wait upon an independently complete QFT ontology has no standing in this proof class. A determinate physical claim already instantiates the kernel. The conventional-facing QFT ledger is essential as public specialization, not as an external authorizer of determinacy.

## D.2 Operative attacks and closures

| Objection                                                             | Standing           | Resolution                                                                                                                                                                               |
|-----------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role positions named their intended occupants                         | Correct            | Version 7 derives slots from target obligations and removes candidate names, source IDs, semantic subjects, and expected-occupant fields.                                                |
| Residues were generated after role predicates passed                  | Correct            | Physical residues are generated before the role architecture is loaded and contain no final role field or alias.                                                                         |
| The eligibility graph was a preconstructed perfect matching           | Correct            | The independent graph has ninety eligible edges; twenty positions and nineteen residues have degree greater than one.                                                                    |
| Joint compatibility did no work                                       | Correct            | Twenty-nine complete preassignments are evaluated and seven are rejected by family-level causal and co-instantiation rules.                                                              |
| The lawful group was trivialized by source IDs and role-bearing names | Correct            | Slot signatures exclude occupant and ordinal data; residue signatures exclude source IDs and display names; only theorem-bearing invariant data enter the group calculation.             |
| Completeness was self-certified                                       | Correct            | Family certificates now contain generated role architecture, residue, quotient, incidence, compatibility, failure, and orbit evidence rather than self-authorizing Boolean declarations. |
| Constructors copied the declared child set                            | Correct limitation | T3 and T14 are stated as registered benchmark decomposition and reconstruction. Full target-generated candidate exhaustion remains assigned to SYM-11.                                   |

| Objection                                                   | Standing                | Resolution                                                                                                                                           |
|-------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support minimality exceeded the audit                       | Correct limitation      | T11 is theorem-relative dependency integrity, explicitly limited to the frozen theorem set and declared relation.                                    |
| Redescription regressions did not prove the general theorem | Correct distinction     | The general quotient theorem is imported and specialized; numerical regressions are evidence of faithful implementation, not its source.             |
| A deleted premise changed output                            | Audit defect, not rival | Deleting a required fact yields an incomplete certificate and no normal form. A theorem counterexample must preserve the complete fixed certificate. |
| Software does not prove CPT or anomalies                    | No standing             | Those are imported proposition-level theorems; software verifies target and certificate propagation.                                                 |

### D.3 Falsification burden

The endpoint fails if a critic supplies a second complete assignment orbit preserving the same target, role obligations, independently formed residue universe, source-certified facts, eligibility relation, joint compatibility, continuation, failure conditions, and lawful quotient. The adversarial suite includes a synthetic faithful second-orbit case and correctly reports theorem failure in that countermodel.

---

## Formalization Boundary

The formal directory records the frozen target, role-position, physical-residue, eligibility, compatibility, automorphism, assignment, orbit, status, support, and theorem interfaces for subsequent Lean development. The next stage must generate definitions from the same source-of-truth registries, instantiate the sixteen family objects, import record-specific physical certificates as explicit assumptions, and prove the finite orbit counts under a pinned toolchain. This publication makes no machine-checked Lean claim.

## Exact Nonclaims

The normal-form theorem does not derive every Standard Model symmetry from no premises, determine numerical couplings or masses, establish Standard Model uniqueness under arbitrary ultraviolet assumptions, classify quantum gravity, or complete the candidate-exhaustion burden assigned to SYM-11. It does not identify representative freedom with unreality, reduce anomalies to one failure type, or promote approximate symmetry to primitive standing. It does not represent the audit software as the source of kernel necessity or imported QFT truth.

Table F.1: Generated record-level nonclaims.

| Record                                 | Blocked inference                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| SYM-00-GAUGE-ARCH-v5                   | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-GAUGE-ARCH-v5                   | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-LOCAL-PROPER-v5           | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-GAUGE-LOCAL-PROPER-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-BOUNDARY-RULE-v5          | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GAUGE-BOUNDARY-RULE-v5          | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-BOUNDARY-RULE-v5          | The record is structurally classified but its unresolved physics is not reported as exact.                                          |
| SYM-00-U1-ASYMPTOTIC-CHARGE-v5         | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-U1-ASYMPTOTIC-CHARGE-v5         | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-LARGE-QUOTIENT-v5         | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-GAUGE-LARGE-QUOTIENT-v5         | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GAUGE-LARGE-QUOTIENT-v5         | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-LARGE-THETA-ACTION-v5     | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GAUGE-LARGE-THETA-ACTION-v5     | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5        | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5        | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GAUGE-TOPOLOGY-ANCHOR-v5        | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 | The changed target is not a same-scope rival to the source target.                                                                  |
| SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GAUGE-TOPOLOGY-TARGET-UPDATE-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |

| Record                              | Blocked inference                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| SYM-00-SU2-GLOBAL-ANOMALY-v5        | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-SU2-GLOBAL-ANOMALY-v5        | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-SU2-GLOBAL-ANOMALY-v5        | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-BRST-COHOMOLOGY-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-LORENTZ-PASSIVE-v5           | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-LORENTZ-PASSIVE-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-POINCARÉ-ACTIVE-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-POINCARÉ-ANCHOR-v5           | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-POINCARÉ-ANCHOR-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-POINCARÉ-NOETHER-WITNESS-v5  | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-CPT-ACTION-v5                | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-CPT-ACTION-v5                | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-CPT-THEOREM-v5               | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-CPT-THEOREM-v5               | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U1A-ABJ-DIAGNOSTIC-v5        | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-U1A-ABJ-DIAGNOSTIC-v5        | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-U1A-ABJ-DIAGNOSTIC-v5        | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U1A-CLASSICAL-ACTION-v5      | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-U1A-CLASSICAL-ACTION-v5      | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-U1A-CLASSICAL-ACTION-v5      | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-U1A-CLASSICAL-ACTION-v5      | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U1A-NONINVERTIBLE-ACTION-v5  | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-U1A-NONINVERTIBLE-ACTION-v5  | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-U1A-NONINVERTIBLE-ACTION-v5  | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5 | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-BPLUSL-NONPERT-DIAGNOSTIC-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-BPLUSL-PERT-ACTION-v5        | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-BPLUSL-PERT-ACTION-v5        | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-BPLUSL-PERT-ACTION-v5        | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U35-ZERO-YUKAWA-ACTION-v5    | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-U35-ZERO-YUKAWA-ACTION-v5    | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U35-ZERO-YUKAWA-ACTION-v5    | The record is structurally classified but its unresolved physics is not reported as exact.                                          |
| SYM-00-CKM-JARLSKOG-DIAGNOSTIC-v5   | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-MIXING-REPHASING-SKIN-v5     | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |

| Record                                    | Blocked inference                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| SYM-00-MIXING-REPHASING-SKIN-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5     | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5     | The changed target is not a same-scope rival to the source target.                                                                  |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5     | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5     | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-PMNS-NEUTRINO-TARGET-UPDATE-v5     | The record is structurally classified but its unresolved physics is not reported as exact.                                          |
| SYM-00-CUSTODIAL-DIAGNOSTIC-v5            | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-CUSTODIAL-DIAGNOSTIC-v5            | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-CUSTODIAL-DIAGNOSTIC-v5            | The record is structurally classified but its unresolved physics is not reported as exact.                                          |
| SYM-00-CUSTODIAL-LIMIT-ACTION-v5          | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-CUSTODIAL-LIMIT-ACTION-v5          | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-CUSTODIAL-LIMIT-ACTION-v5          | The record is structurally classified but its unresolved physics is not reported as exact.                                          |
| SYM-00-EW-LOCAL-GAUGE-SKIN-v5             | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-EW-LOCAL-GAUGE-SKIN-v5             | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-EW-VACUUM-FIXATION-v5              | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-EW-VACUUM-FIXATION-v5              | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GOLDSTONE-REPRESENTATIVE-v5        | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-GOLDSTONE-REPRESENTATIVE-v5        | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-U1EM-RESIDUAL-ACTION-v5            | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5       | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5       | The changed target is not a same-scope rival to the source target.                                                                  |
| SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5       | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GLOBAL-FORM-TARGET-UPDATE-v5       | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5           | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5           | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-GLOBAL-FORM-Z6-ANCHOR-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-LINE-SPECTRUM-STRUCTURE-v5         | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-LINE-SPECTRUM-STRUCTURE-v5         | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-LINE-SPECTRUM-STRUCTURE-v5         | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-HIGHER-FLAVOR-BACKGROUND-ACTION-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5         | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |

| Record                             | Blocked inference                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5  | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5  | No numerical coupling, mass, mixing-angle, or scale value follows without a separate numerical certificate.                         |
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5  | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-HIGHER-FLAVOR-EXTENSION-v5  | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-ONEFORM-ACTION-v5           | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-ONEFORM-ACTION-v5           | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-EFT-THRESHOLD-MATCHING-v5   | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-EFT-THRESHOLD-MATCHING-v5   | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-EFT-THRESHOLD-MATCHING-v5   | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-FIXED-POINT-SCALE-ACTION-v5 | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-FIXED-POINT-SCALE-ACTION-v5 | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-FIXED-POINT-SCALE-ACTION-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-RG-PHYSICAL-FLOW-v5         | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-RG-PHYSICAL-FLOW-v5         | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-RG-SCALE-COORDINATE-v5      | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-RG-SCALE-COORDINATE-v5      | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-RG-SCHEME-REDESCRIPTION-v5  | Representative equivalence does not imply that the licensing architecture or invariant observable content is unreal or dispensable. |
| SYM-00-RG-SCHEME-REDESCRIPTION-v5  | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-SCALE-CLASSICAL-ACTION-v5   | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-SCALE-CLASSICAL-ACTION-v5   | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-SCALE-CLASSICAL-ACTION-v5   | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5 | An obstruction is typed by locus; it does not erase every meaningful action, remnant, or matching consequence.                      |
| SYM-00-TRACE-ANOMALY-DIAGNOSTIC-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-HQET-SPIN-FLAVOR-ACTION-v5  | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-HQET-SPIN-FLAVOR-ACTION-v5  | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-HQET-SPIN-FLAVOR-ACTION-v5  | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |
| SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5 | A regime-limited symmetry or transport relation is not promoted to an all-scale primitive carrier.                                  |
| SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5 | The conclusion is restricted to the explicitly named controlled target variant.                                                     |
| SYM-00-ISOSPIN-EFFECTIVE-ACTION-v5 | The record does not establish full Standard Model symmetry exhaustion or uniqueness under arbitrary ultraviolet premises.           |

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